

Determinacy in $L(\mathbb{R})$

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0 Introduction

- I. In these notes, we prove that, under the suitable large cardinal assumptions, model $L(\mathbb{R})$ satisfies Axiom of Determinacy. We do so by proving to theorems which are of interest in their own right. First, we prove that a *derived model* satisfies Axiom of Determinacy. Second, we prove that $L(\mathbb{R})$ can be (generically) embedded into the derived model.
- II. Here, by a *derived model*, we mean the model $L(\mathbb{R}^*)$, where $\mathbb{R}^*$ are reals obtained by symmetrically collapsing a limit of Woodin cardinals. Usually, the derived model is somewhat bigger, i.e. $L(\mathbb{R}^*, \text{Hom}^*)$, where Hom^* is a certain collection of sets of reals (cf. [Steel, 2003]). The two proofs have essentially the same structure, but we opted for the technically simpler one.
- III. The proof of the derived model theorem has been presented to us by G. Sargsyan. Other than that, we heavily rely on [Larson et al., 2004] and [Steel, 2003].
- IV. These notes are mostly self-contained, with following exceptions. First, Martin-Steel Transfer Theorem is cited without a proof. We presented a proof of this theorem in [Kasum, 2021a]. Second, we use the stationary tower as the main technical tool; we gave an exposition on it in [Kasum, 2021b]. In the same notes,

one can find proof that the Chang's model of the ground can be generically embedded into the Chang's model obtained after collapsing a Woodin limit of Woodin cardinals. Construction of this embedding is highly analogous to the construction of the embedding between $L(\mathbb{R})$ and $L(\mathbb{R}^*)$ presented here.

1 Regularity Properties

1.1 Homogeneity

- I. **Definition.** A *tower of measures with support Z* is a sequence $(\mu_n : n < \omega)$ such that μ_m is a measure on Z^m and μ_n projects on μ_m for $m \leq n < \omega$.
- II. If $f : Z^m \rightarrow V$ and $n \geq m$, we can define $f^n : Z^n \rightarrow V$ by simply putting $f^n(s) := f(s \restriction m)$ for $s \in Z^n$. For $m \leq n < \omega$, we define

$$j_{m,n}^\mu : \text{Ult}(V, \mu_m) \rightarrow \text{Ult}(V, \mu_n)$$

by setting $j_{m,n}^\mu([f]_{\mu_m}) := [f^n]_{\mu_n}$. This mapping is correctly defined and elementary.

- III. **Definition.** Let μ be a tower of measures. The *ultrapower* $\text{Ult}(V, \mu)$ of V via μ is defined as the direct limit of the system.

$$(\text{Ult}(V, \mu_m), j_{m,n}^\mu : m \leq n < \omega).$$

The Loś's theorem holds in its expected form. The *ultrapower embedding* is given by

$$j_\mu := j_{0,\infty}^\mu : V = \text{Ult}(V, \mu_0) \prec \text{Ult}(V, \mu).$$

- IV. $\text{Ult}(V, \mu)$ can also be described as $(D, =^D, \in^D) / =^D$, where:

$$\begin{aligned} D &:= \{f : Z^m \rightarrow V : m < \omega\}, \\ f =^D g &: \iff \{u \in Z^{\max\{m,n\}} : f(u \restriction m) = g(u \restriction n)\}, \\ f \in^D g &: \iff \{u \in Z^{\max\{m,n\}} : f(u \restriction m) \in g(u \restriction n)\} \\ & \quad (f : Z^m \rightarrow V, g : Z^n \rightarrow V). \end{aligned}$$

- V. **Definition.** A tower μ of measures with support Z is *complete* if for every $(A_n : n < \omega)$ with $A_n \in \mu_n$, there is $f \in Z^\omega$ such that

$$(\forall n < \omega)(f \restriction n \in A_n).$$

- VI. **Proposition.** $\text{Ult}(V, \mu)$ is well-founded if and only if μ is complete.

Proof.

Implication $(\Rightarrow)$

- 1° Let $(A_n : n < \omega)$ be such that $\mu_n(A_n) = 1$ for every $n < \omega$.
- 2° We may assume that $\{u \upharpoonright n : u \in A_{n+1}\} \subseteq A_n$ for every $n < \omega$.
- 3° Let $j : V \prec \text{Ult}(V, \mu)$ be the ultrapower embedding.
- 4° Since $\mu_n(A_n) = 1$, we have $[\text{id}_{Z^n}]_\mu \in j(A_n)$.
- 5° $T := \bigcup_{n < \omega} A_n$ is a tree on Z in V .
- 6° We have $[\text{id}_{Z^n}]_\mu \triangleleft [\text{id}_{Z^{n+1}}]_\mu$ for every n .
- 7° By 4°, we see that $\bigcup_{n < \omega} [\text{id}_{Z^n}]_\mu$ is a branch through $j(T)$ in V .
- 8° By absoluteness, there is a branch through $j(T)$ in $\text{Ult}(V, \mu)$.
- 9° By elementarity, there is a branch f through T in V .
- 10° f is as required for $(A_n : n < \omega)$.

Implication ($\Leftarrow$)

- 11° Suppose that $\text{Ult}(V, \mu)$ is ill-founded.
- 12° Since it is the direct limit of $(\text{Ult}(V, \mu_m), j_{m,n}^\mu)_{m,n}$, there are $f_m : Z^m \rightarrow \text{Ord}$ such that

$$j_{m,m+1}([f_m]) > [f_{m+1}]$$

for every m .

- 13° Set $A_{m+1} := \{s \in Z^{m+1} : f_{m+1}(s) < f_m(s \upharpoonright m)\} \in \mu_{m+1}$.
- 14° There is $f \in Z^\omega$ such that $f \upharpoonright (m+1) \in A_{m+1}$ for every m .
- 15° Then $(f_{m+1}(f \upharpoonright (m+1)) : m < \omega)$ is a decreasing sequence of ordinals, which is a contradiction.

□

VII. Definition. A κ -homogeneity system on X with support Z is a family $\mu = (\mu_s : s \in X^{<\omega})$ satisfying:

- a. μ_s is a measure on $Z^{|s|}$;
- b. μ_s is $\max\{|X|^+, \kappa\}$ -complete;
- c. μ_t projects to μ_s for $s \triangleleft t$.

For $x \in X^\omega$, we denote by μ_x the tower $(\mu_{x \upharpoonright n} : n < \omega)$.

VIII. Remark. If $\kappa = \omega_1$, the parameter κ is omitted from the terminology.

IX. For μ a homogeneity system, we denote $M_s^\mu := \text{Ult}(V, \mu_s)$ and $j_{s,t}^\mu : M_s^\mu \prec M_t^\mu$ the embedding $j_{s,t}^{\mu_x}$ of some/any branch $x \triangleright t \triangleright s$. The limit models and embeddings are denoted by M_x^μ and $j_{s,x}^\mu$ ($s \triangleleft x \in X^\omega$).

X. Definition. For a homogeneity system μ , we define the *set of well-founded branches of μ* to be

$$S_\mu := \{x \in X^\omega : \text{Ult}(V, \mu_x) \text{ is well-founded}\}.$$

A set $A \subseteq X^\omega$ is said to be κ -homogeneous if there is a κ -homogeneity system μ on X such that $A = S_\mu$.

XI. **Definition.** Let T be a tree on $X \times Z$. A κ -homogeneity system for T is a κ -homogeneity system μ on X with support Z such that:

- a. for every $s \in X^{<\omega}$, we have $\mu_s(T_s) = 1$;
- b. $p[T] \subseteq S_\mu$.

T is κ -homogeneous if there exists a κ -homogeneity system for T .

XII. **Proposition.** If μ is a κ -homogeneity system for T , then in fact $p[T] = S_\mu$.

Proof.

- 1° We only need to verify $S_\mu \subseteq p[T]$.
- 2° Let $x \in X^\omega$ be such that μ_x is complete.
- 3° For every $n < \omega$, we have $\mu_{x \upharpoonright n}(T_{x \upharpoonright n}) = 1$.
- 4° There there is $f \in Z^\omega$ such that $f \upharpoonright n \in T_{x \upharpoonright n}$ for every $n < \omega$.
- 5° f witnesses $x \in p[T]$.

□

XIII. **Definition.** A set $A \subseteq X^\omega$ is κ -homogeneously Suslin if there is a κ -homogeneous tree T with $A = p[T]$.

XIV. **Proposition.** The following are equivalent for $A \subseteq X^\omega$:

- a. A is κ -homogeneous;
- b. A is κ -homogeneously Suslin.

Proof.

Implication (b $\Rightarrow$ a)

- 1° It follows immediately from 1.1-XII.

Implication (a $\Rightarrow$ b)

- 2° Let $(\mu_s : s \in X^{<\omega})$ be a κ -homogeneity system on X with support Z such that $A = S_\mu$.
- 3° We need to find T on $X \times Z$ such that (1) $p[T] = A$ and (2) $\mu_s(T_s) = 1$ for every s .
- 4° If $x \in A^\complement$, then there are $(B_n^x : n < \omega)$ such that
 - a. $\mu_{x \upharpoonright n}(B_n^x) = 1$ for every n ,
 - b. there is no $z \in Z^\omega$ satisfying $(\forall n < \omega)(z \upharpoonright n \in B_n^x)$.
- 5° For $x \in A$, we set $B_n^x := Z^n$.
- 6° Since additivity of μ_s is a measurable cardinal¹ $> |X|$, we have that

$$\mu_s \left(\bigcap \{B_{|s|}^x : s \triangleleft x \in X^\omega\} \right) = 1.$$

¹or the measure is a principal ultrafilter

7° $T'_s := \bigcap \{B_{|s|}^x : s \triangleleft x \in X^\omega\}$ is of measure-1 for all s .

8° By recursion on length of $s \in X^{<\omega}$, set:

a. $T''_0 := T'_0$

b. $T''_s := T'_s \cap \{u \in Z^{|s|} : u \upharpoonright (|s| - 1) \in T''_{s \upharpoonright (|s| - 1)}\}$

9° We have that every T''_s is of measure-1 and

$$(\forall u \in T''_s)(u \upharpoonright (|s| - 1) \in T''_{s \upharpoonright (|s| - 1)}).$$

10° $T := \bigcup_{s \in X^{<\omega}} (\{s\} \times T''_s)$

11° We have that T is a tree on $X \times Z$ and $T_s = T''_s$ is of measure one for every s .

12° It remains to show that $p[T] = A$.

13° $A \subseteq p[T]$:

1' Let $x \in A$.

2' Then $\text{Ult}(V, \mu_x)$ is well-founded.

3' Then there is a fiber $z \in Z^\omega$ through $(T_s : s \triangleleft x)$.

4' z witnesses $x \in p[T]$.

14° $p[T] \subseteq A$:

1' Assume $x \in A^c$.

2' Then there is no fiber through $(B_n^x : n < \omega)$.

3' Since $T_s \subseteq B_{|s|}^x$ for $s \triangleleft x$, there is no fiber through $(T_s : s \triangleleft x)$.

4' Thus, $x \notin p[T]$.

□

XV. Proposition. Let $f : X^\omega \rightarrow Y^\omega$ be continuous and suppose that $B \subseteq Y^\omega$ is κ -homogeneous. Then $f^{-1}[B] \subseteq X^\omega$ is also κ -homogeneous.

Proof.

1° Function $g : X^{<\omega} \rightarrow Y^{<\omega}$ is defined in the following manner.

1' Take $s \in X^{<\omega}$.

2' There is maximal $n \leq |s|$ such that the value $f(x) \upharpoonright n$ is constant for every $x \triangleright s$.

3' We define $g(s) := f(x) \upharpoonright n$ for some/any $x \triangleright s$.

2° g has the following properties for $s, t \in X^{<\omega}$ and $x \in X^\omega$:

a. $s \triangleleft t \implies g(s) \trianglelefteq g(t)$

b. $|g(s)| - |g(s \upharpoonright (|s| - 1))| \leq 1$

c. $\forall s \forall n \exists t (s \triangleleft t \wedge |g(t)| \geq n)$;

d. $f(x) = \bigcup_{n < \omega} g(x \upharpoonright n)$.

3° Let $\nu = (\nu_s : s \in Y^{<\omega})$ be a κ -homogeneity system with support Z such that $B = S_\nu$.

4° For $s \triangleleft t \in X^{<\omega}$, we define

$$k_{s,t} := j_{g(s),g(t)}^\nu : \text{Ult}(V, \nu_{g(s)}) \prec \text{Ult}(V, \nu_{g(t)}).$$

5° We may assume w.l.o.g. that $0 \in Z$.

6° For $s \in X^{<\omega}$, we define tuples $u_s \in j_s(Z)^{|s|}$ by induction on $|s|$.

1' $u_\emptyset := \emptyset$

2' Suppose that the stage $|s| = n$ has been handled and let us consider the stage $|s| = n + 1$.

3' If $g(s) \triangleright g(s \upharpoonright n)$, let z_s be the last coordinate of $[\text{id}_{Z|g(s)|}]_{\nu_{g(s)}}$ and define $u_s := k_{s \upharpoonright n, s}(u_{s \upharpoonright n}) \frown z_s$.

4' Otherwise (if $g(s) = g(s \upharpoonright n)$), we define $u_s := u_{s \upharpoonright n} \frown 0$.

7° Let μ_s be the measure on $Z^{|s|}$ derived from j_s at u_s .

8° $\mu := (\mu_s : s \in X^{<\omega})$ is a κ -homogeneity system.

9° Also, $\text{Ult}(V, \mu_s) = \text{Ult}(V, \nu_{g(s)})$ for every s , and consequently, $S_\mu = f^{-1}[S_\nu]$.

10° Thus, $f^{-1}[B] = S_\mu$ is κ -homogeneously Suslin.

□

XVI. *Raison d'être* for the notion of homogeneity is the following theorem.

XVII. **Theorem.** (Martin) If $A \subseteq X^\omega$ is homogeneous, then A is determined.

Proof.

1° Let T be a homogeneous tree on $X \times Z$ projecting to A and let μ be a homogeneity system for T .

2° We consider game G

$$\begin{array}{c|cccc} \text{I} & x_0 & x_2 & & \cdots \\ \hline \text{II} & & x_1 & x_3 & \cdots \end{array}$$

where $x_i \in X$ and I wins if and only if $(x_i)_i \in A$.

3° Let us consider auxiliary game G^*

$$\begin{array}{c|cccc} \text{I} & x_0, z_0 & x_2, z_1, z_2 & x_4, z_3, z_4 & \cdots \\ \hline \text{II} & & x_1 & x_3 & \cdots \end{array}$$

where $x_i \in X$, $z_i \in Z$, and I wins if and only if

$$(z_i : i < n) \in T_{(x_i : i < n)} \tag{1}$$

for every $n < \omega$.

4° G^* is obviously closed for I, so one of the players have a winning strategy.

- 5° If I wins G^* , then they can use the same strategy, after omitting z_i 's, to win G . The reason this modification is winning in G is because (1) ensures $(x_i : i < \omega) \in p[T] = A$.
- 6° Let us assume now that II has a winning strategy σ^* in G^* and let us describe their winning strategy σ in G .
- 7° Suppose recursively that we are given a partial play

I	x_0	x_2	$\cdots$	x_{2n}
II		x_1	x_3	$\cdots$

of G played according to σ and let us determined what should be the next move x_{2n+1} according to σ .

- 8° To apply σ^* , we need to know moves $z_0, \dots, z_{2n}$ of I in G^* which are not given.
- 9° However, for every such tuple $t \in Z^{2n+1}$, there is a move $x_{2n+1}(t) \in X$ given by σ^* based on $s := (x_i : i \leq 2n)$ and t .
- 10° Function $t \mapsto x_{2n+1}(t)$ is constant on a set $C_s \subseteq T_s$ of μ_s -measure one, with some (unique) value $\sigma(s)$. (This value is the next move for II according to σ .)
- 11° **Claim.** σ is winning for II in G .

Proof.

- 1' Let x be a play of G played according to σ and let us assume $x \in A$.
- 2' Then $(\mu_s : s \triangleleft x)$ is complete.
- 3' Since $(C_{s \upharpoonright (2n+1)} : n < \omega)$ is a sequence of measure-one sets, there is $z \in Z^\omega$ such that $z \upharpoonright (2n+1) \in C_{s \upharpoonright (2n+1)}$ for every $n < \omega$.
- 4' x and z now determine a move according to σ^* such that $z \upharpoonright n \in T_{x \upharpoonright n}$ for every $n < \omega$.
- 5' This play is won by I, but played according to σ^* , which is a contradiction.

□

□

1.2 Weak Homogeneity

- I. **Definition.** A *weak κ -homogeneity system on X with support Z* is a κ -homogeneity system on $X \times \omega$ with support Z . The *projection of the set of well-founded branches of μ* is the set

$$W_\mu := \{x \in X^\omega : (\exists y \in \mathbb{R})(\text{Ult}(V, \mu_{x,y}) \text{ is well-founded})\} = \exists^\mathbb{R} S_\mu.$$

- II. **Definition.** A set $A \subseteq X^\omega$ is *weakly κ -homogeneous* if there exists a weak κ -homogeneity system μ on X such that $A = W_\mu$.

III. **Proposition.** The following are equivalent for $A \subseteq X^\omega$:

- a. A is weakly κ -homogeneous;
- b. there exists $B \subseteq X^\omega \times \mathbb{R}$ that is κ -homogeneous and that projects to A .

Proof. Obvious. □

IV. **Definition.** A *weak κ -homogeneity system* for a tree T on $X \times Z$ is a weak κ -homogeneity system μ on X with support Z satisfying:

- a. for every $(s, t) \in (X \times \omega)^{<\omega}$, we have $\mu_{s,t}(T_s) = 1$;
- b. $p[T] \subseteq W_\mu$.

T is a *weakly κ -homogeneous tree* if there exists a weak κ -homogeneity system for T .

V. **Proposition.** If μ is a weak homogeneity system for T , then in fact $p[T] = W_\mu$.

Proof.

- 1° We need to show $(\supseteq)$.
- 2° Let (x, y) be such that $\text{Ult}(V, \mu_{x,y})$ is well-founded. We want to show that $x \in p[T]$.
- 3° For every $n < \omega$, we have $\mu_{x \upharpoonright n, y \upharpoonright n}(T_{x \upharpoonright n}) = 1$.
- 4° Since $\mu_{x,y}$ is a complete tower, there is thread $z \in Z^\omega$ through $(T_{x \upharpoonright n} : n < \omega)$.
- 5° z witnesses $x \in p[T]$.

□

VI. **Definition.** A set $A \subseteq X^\omega$ is *weakly κ -homogeneously Suslin* if there is a weakly κ -homogeneous tree T such that $A = p[T]$.

VII. **Proposition.** The following are equivalent for a set $A \subseteq X^\omega$:

- a. A is weakly κ -homogeneous;
- b. A is weakly κ -homogeneously Suslin;
- c. A is the projection of a κ -homogeneous set.

Proof.

Implication $(b \Rightarrow a)$

- 1° It follows from 1.2-V.

Implication $(a \Rightarrow b)$

- 2° By 1.2-III, for some Z , we have a κ -homogeneous tree T on $(X \times \omega) \times Z$ such that $A = p[p[T]]$.
 - 3° Let $(\mu_{s,t} : (s,t) \in (X \times \omega)^{<\omega})$ be a κ -homogeneity system for T .
 - 4° Tree T can be seen as a tree U on $X \times (\omega \times Z)$.
 - 5° We have $p[U] = A$.
 - 6° For $(s,t) \in (X \times \omega)^{<\omega}$, we define $\nu_{s,t}$ as the measure on $\omega^{|s|} \times Z^{|s|}$ which to a set $C \subseteq \omega^{|s|} \times Z^{|s|}$, assigns measure $\mu_{s,t}(\{u \in Z^{|s|} : (t,u) \in C\})$.
 - 7° It is easily seen that $(\nu_{s,t} : (s,t) \in (X \times \omega)^{<\omega})$ witnesses that U is weakly κ -homogeneous.
- Equivalence (a $\Leftrightarrow$ c)**
- 8° This is proposition 1.2-III.

□

VIII. Weak homogeneity of a tree on $\omega \times Z$ can be reformulated in the following manner.

IX. **Proposition.** Let T be a pruned² tree on $\omega \times Z$ and let $\kappa > \omega$. Then T is weakly κ -homogeneous if and only if there is a countable set σ satisfying:

- a. every element σ is a κ -complete measure on some Z^n ;
- b. for every $x \in p[T]$, there exists a complete tower $\mu \in \sigma^\omega$, such that for every $s \triangleleft x$, we have $\mu_{|s|}(T_s) = 1$.

Proof.

Implication ($\Rightarrow$)

- 1° If μ is weak κ -homogeneity system for T , then we can take $\sigma := \{\mu_{x,y} : (x,y) \in (\omega \times \omega)^{<\omega}\}$.

Implication ($\Leftarrow$)

- 2° We may assume that σ is closed for projections of measures.
- 3° A weak κ -homogeneity system $(\mu_{s,t} : (s,t) \in (\omega \times \omega)^{<\omega})$ is defined by recursion on $|s|$.
- 4° The case $|s| = 0$ is obvious.
- 5° Assume that the case $|s| = n$ has been done and let us handle $|s| = n + 1$.
- 6° For any $t \in \omega^n$, we have two case.
- 7° **The first case** is when there is no measure $\nu \in \sigma$ on Z^{n+1} that projects to $\mu_{s \upharpoonright n,t}$ and satisfies $\nu(T_s) = 1$.
- 8° **Claim.** There exists some measure ν on Z^{n+1} that projects to $\mu_{s \upharpoonright n,t}$ and satisfies $\nu(T_s) = 1$.

²I.e. T has no $\triangleleft$ -maximal elements.

Proof.

- 1' Consider $j : V \prec \text{Ult}(V, \mu_{s \upharpoonright n, t})$. Measure $\mu_{s \upharpoonright n, t}$ is derived from j at $u := [\text{id}_{Z^n}] \in T_{s \upharpoonright n}$.
- 2' Every element of $T_{s \upharpoonright n}$ is an initial segment of some element of T_s .
- 3' Consequently, $u \in j(T_{s \upharpoonright n})$ can be extended to some $u' \in j(T_s)$.
- 4' Let ν be the measure on Z^{n+1} derived from j at u' .
- 5' ν is as required.

□

- 9° Set $\mu_{s, t \smallfrown i} := \nu$ for every $i < \omega$.
- 10° **The second case** is when there is at least one measure $\nu \in \sigma$ on Z^{n+1} that projects to $\mu_{s \upharpoonright n, t}$ and satisfies $\nu(T_s) = 1$.
- 11° Then all such measures can be enumerated as $(\mu_{s, t \smallfrown i} : i < \omega)$.
- 12° This finishes the recursion. It only remains to verify $p[T] \subseteq W_\mu$.
- 13° Let $x \in p[T]$ be arbitrary.
- 14° Then there is a tower $\nu \in \sigma^\omega$ such that $\nu_{|s|}(T_s) = 1$ for every $s \triangleleft x$.
- 15° A simple induction now shows that for every $n < \omega$, there exists $t_n \in \omega^{<\omega}$ extending all the previous t_i 's and satisfying $\mu_{x \upharpoonright n, t_n} = \nu_n$.
- 16° $y := \bigcup_{n < \omega} t_n$ has property that $\mu_{x, y} = \nu$.
- 17° Thus, $\mu_{x, y}$ is complete, implying $x \in W_\mu$.

□

X. **Lemma.** Suppose that tree T on $X \times \omega \times Z$ is weakly κ -homogeneous. Then tree $U := \{(s, (t, u)) : (s, t, u) \in T\}$ is weakly κ -homogeneous.

Proof.

- 1° Let $(\mu_{s, t, v} : (s, t, v) \in (X \times \omega \times \omega)^{<\omega})$ be a weak κ -homogeneity system for T .
- 2° Let $(\text{decode}_0, \text{decode}_1) : \omega^{<\omega} \rightarrow (\omega \times \omega)^{<\omega}$ be a bijection that preserves lengths.
- 3° For $(s, t) \in (X \times \omega)^{<\omega}$, we define measure $\nu_{s, t}$ on $\omega^{|s|} \times Z^{|s|}$ by setting
$$\nu_{s, t}(C) := \mu_{s, \text{decode}_0(t), \text{decode}_1(t)}(\{u \in Z^{|s|} : (\text{decode}_0(t), u) \in C\}).$$
- 4° We get a weak κ -homogeneity system $(\nu_{s, t} : (s, t) \in (X \times \omega)^{<\omega})$ for U .

□

XI. **Proposition.**

- a. If $A, B \subseteq X^\omega$ are weakly κ -homogeneous, then so is $A \cup B$.

- b. If $A \subseteq X^\omega \times \mathbb{R}$ is weakly κ -homogeneous, then so is $\exists^\mathbb{R} A$.
- c. If $f : X^\omega \rightarrow Y^\omega$ is continuous and $B \subseteq Y^\omega$ is weakly κ -homogeneous, then $f^{-1}[B]$ is weakly κ -homogeneous as well.

Proof.

1° **Part a.**

2° Suppose $A = W_\mu$ and $B = W_\nu$, where

$$\mu = (\mu_{s,t} : s \in (X \times \omega)^{<\omega}), \quad \nu = (\nu_{s,t} : (s,t) \in (X \times \omega)^{<\omega})$$

are weak κ -homogeneity systems. We may assume w.l.o.g. that all the measures have the same support.

3° Let

$$t'(n) := \begin{cases} t(0)/2, & n = 0, 2|t(0) \\ t(0)/2 - 1, & n = 0, 2 \nmid t(0) \\ t(n), & n > 0 \end{cases}$$

for $t \in \omega^{<\omega} - \{\emptyset\}$.

4° Let

$$\xi_{s,t} := \begin{cases} \mu_{\emptyset,\emptyset} = \nu_{\emptyset,\emptyset}, & |t| = 0 \\ \mu_{s,t'}, & |t| > 0, 2|t(0) \\ \nu_{s,t'}, & |t| > 0, 2 \nmid t(0) \end{cases}$$

for $(s,t) \in (X \times \omega)^{<\omega}$.

5° $\xi := (\xi_{s,t})_{s,t}$ is a weak κ -homogeneity system and $A \cup B = W_\xi$.

6° **Part b.**

7° It follows from the previous lemma.

8° **Part c.**

9° The proof is the obvious modification of proof [1.1-XV](#).

□

1.3 Absolute Complementation

I. **Proposition.** Let T be a tree on $X \times Z$, let U be a tree on $X \times W$, and suppose that $p[T] \cap p[U] = \emptyset$. Then for every V -generic G , we have

$$V[G] \models p[T] \cap p[U] = \emptyset.$$

Proof.

1° Assume otherwise.

2° Let $S := \{(s,u,v) : (s,u) \in T, (s,v) \in U\}$.

3° Since $p[T] \cap p[U] \neq \emptyset$ in $V[G]$, we have that S is ill-founded.

4° By absoluteness, S is ill-founded in V .

5° This implies $V \models p[T] \cap p[U] \neq \emptyset$, which is a contradiction.

□

II. If $p[T] \sqcup p[U] = X^\omega$, then in $V[G]$, we have $p[T] \sqcup p[U] \subseteq X^\omega$. We introduce the following notion.

III. **Definition.** Trees T on $X \times Z$ and U on $X \times W$ are said to be κ -*absolutely complementing* if for every poset $\mathbb{P}$ of cardinality $< \kappa$, we have

$$V^\mathbb{P} \models p[T] \sqcup p[U] = X^\omega.$$

Tree T is κ -*absolutely complemented* if there exists a tree U such that T and U are κ -absolutely complementing.

IV. **Proposition.** If T and T' are κ -absolutely complemented and $p[T] = p[T']$, then for every $\mathbb{P}$ with $|\mathbb{P}| < \kappa$, we have

$$V^\mathbb{P} \models p[T] = p[T'].$$

Proof.

1° Let $U^{(i)}$ witness that $T^{(i)}$ is κ -absolutely complemented ($i = 0, 1$).

2° Let $\mathbb{P}$ be a poset with $|\mathbb{P}| < \kappa$, let $G \rightsquigarrow V^\mathbb{P}$, and let us work in $V[G]$.

3° Assume towards contradiction that $p[T] \neq p[T']$.

4° We may assume w.l.o.g. that $p[T] \not\subseteq p[T']$.

5° This means that $p[T] \cap p[U'] \neq \emptyset$.

6° This implies $V \models p[T] \cap p[U'] \neq \emptyset$.

7° Since $V \models p[U'] = p[T']^\mathbb{G} = p[T]^\mathbb{G}$, we have reached a contradiction.

□

V. Thus, if $A \subseteq X^\omega$ is the projection of a κ -absolutely complemented tree, it has a canonical extension to any $< \kappa$ -generic extension.

VI. **Definition.** Set $A \subseteq X^\omega$ is κ -*universally Baire* if it is a projection of a κ -absolutely complemented tree. For a κ -universally Baire set A and $G \rightsquigarrow_{< \kappa} V$, we define $A^G \subseteq (X^\omega)^{V[G]}$ as the projection $p[T]^{V[G]}$ of some/any κ -absolutely complemented tree $T \in V$ with $p[T]^V = A$.

VII. We want to show that weakly κ -homogeneous sets are κ -universally Baire. We will do that by showing that weakly κ -homogeneous *trees* are κ -absolutely complemented. The absolute complement is the so-called *Martin-Solovay tree*. We first study the case of κ -homogeneous trees, and then we modify the argument to get the result for weak κ -homogeneity.

VIII. Let $j : V \prec M$ with $\text{crit } j = \kappa$, let $\mathbb{P} \in V_\kappa$ and let $G \rightsquigarrow V^\mathbb{P}$. Then

$$j^G : V[G] \rightarrow M[G] : \dot{x}^G \mapsto j(\dot{x})^G$$

is correctly defined, elementary, and $j^G \upharpoonright V = j$.

IX. Let μ be a κ -additive measure on Z and let

$$j := j_\mu : V \prec \text{Ult}(V, \mu) =: M.$$

We have $z_0 := [\text{id}_Z] \in j(Z)$ and for $C \subseteq Z$, we have

$$C \in \mu \iff [\text{id}_Z] \in j(C).$$

Let $\mathbb{P} \in V_\kappa$ and let $G \rightsquigarrow V^\mathbb{P}$. We can derive a measure μ^G on Z at z_0 in $V[G]$ from j^G : for every $C \subseteq Z$, we have

$$C \in \mu^G \iff z_0 \in j^G(C).$$

X. **Lemma.** For a κ -complete measure μ on Z , we have the following.

- a. μ^G is κ -complete.
- b. We have $\mu^G \cap P^V(Z) = \mu$.
- c. For $C \in P^{V[G]}(Z)$, we have

$$\mu^G(C) = 1 \iff (\exists B \in P(C) \cap V)(\mu(B) = 1).$$

- d. If $f : Z \rightarrow V$ is in $V[G]$, then there is $g : Z \rightarrow V$ in V such that

$$\{z \in Z : f(z) = g(z)\} \in \mu^G.$$

Proof. Routine. □

XI. **Lemma.** Let $(\mu_s : s \in X^{<\omega})$ be a κ -homogeneity system and let G be $<\kappa$ -generic over V . Then $(\mu_s^G : s \in X^{<\omega})$ is κ -homogeneity system in $V[G]$.

Proof. We only need to check that μ_t^G projects to μ_s^G for $s \triangleleft t$, but this is obvious from c. □

XII. **Definition.** Let μ be a homogeneity system on X . For a given ordinal θ , we define the *Martin-Solovay tree* $\text{mst}(\mu, \theta)$ on $X \times \text{Ord}$ as consisting of all $(s, t) \in (X \times \text{Ord})^n$ satisfying $t_0 < \theta$ and

$$(\forall i < n-1)(j_{s \upharpoonright i, s \upharpoonright (i+1)}^\mu(t_i) > t_{i+1}).$$

XIII. **Remark.** For some $x \in X^\omega$, the tree $\text{mst}(\mu, \theta)_x$ searches for the proof that $\text{Ult}(V, \mu_x)$ is ill-founded below the image of θ .

XIV. **Proposition.** For $\theta \geq |Z|^+$, we have $p[\text{mst}(\mu, \theta)] = S_\mu^\mathbb{G}$.

Proof.

Inclusion ($\subseteq$).

- 1° Let $(x, y) \in \text{mst}(\mu, \theta)$ be arbitrary.
- 2° Then $(j_{x \upharpoonright i, x}^\mu(y_i) : i < \omega)$ is a decreasing sequence of ordinals in M_x^μ .

Inclusion ($\supseteq$).

- 3° Let $x \in X^\omega$ be such that M_x^μ is ill-founded.
- 4° There are $(f_n : n < \omega)$ such that $f_n : Z^n \rightarrow \text{Ord}$ and

$$(\forall n < \omega)(j_{x \upharpoonright n, x \upharpoonright (n+1)}^{\mu_x}([f_n]_{\mu_x \upharpoonright n}) > [f_{n+1}]_{\mu_x \upharpoonright (n+1)}).$$

- 5° Let $\pi : \bigcup_{n < \omega} \text{im } f_n \rightarrow \alpha$ be the transitive collapse. We have $\alpha < |Z|^+$.
- 6° We have that $([\pi \circ f_n]_{\mu_x \upharpoonright n} : n < \omega)$ witnesses $x \in p[\text{mst}(\mu, \theta)]$.

□

XV. **Proposition.** Suppose that:

- $|Z| \geq \kappa$
- T is a tree on $X \times Z$ in V ;
- μ is a κ -homogeneity system for T ;
- $\theta \geq |Z|^+$
- $U := \text{mst}(\mu, \theta)$.

Then:

- a. U is κ -absolute complement for T ;
- b. if G be $<\kappa$ -generic over V , then μ^G is a κ -homogeneity system for T in $V[G]$.

Proof.

- 1° Let G be arbitrary $<\kappa$ -generic over V . We want to show that in $V[G]$:
 - a. $p[U] = p[T]^\mathbb{G}$;
 - b. $p[T] = \{x \in X^\omega : \text{Ult}(V[G], \mu_x^G) \text{ is well-founded}\}$.
- 2° ($\subseteq$) in **a** is obvious since $V \models p[T] \cap p[U] = \emptyset$.
- 3° ($\supseteq$) of **b**:
 - 1' Suppose that μ_x^G is complete.
 - 2' We have $\mu^G(T_s) = \mu(T_s) = 1$ for every $s \triangleleft x$.
 - 3' It follows that there is a fiber z through $(T_s : s \triangleleft x)$.

- 4' z witnesses $x \in p[T]$.
- 4° ($\supseteq$) of **a**:
- 1' Suppose $x \in p[U]^{\mathbb{C}}$.
- 2' We have $U = \text{mst}(\mu^G, \theta)$.
- 3' By 1.3-XIV, we have that $\text{Ult}(V[G], \mu_x^G)$ is well-founded.
- 4' Thus, $x \in p[T]$ by 3°.
- 5° We have established **a**.
- 6° ($\subseteq$) of **b**:
- 1' Let $x \in p[T]$.
- 2' By **a**, we have $x \in p[U]^{\mathbb{C}}$.
- 3' Since $U = \text{mst}(\mu^G, \theta)$, we have that $\text{Ult}(V[G], \mu_x^G)$ is well-founded by 1.3-XIV.
- 7° This finishes the proof.

□

XVI. We now modify the above argument to handle the case of weak homogeneity. Let us fix the usual coding $\mathbf{t} : \omega \rightarrow \omega^{<\omega}$ and set $l_i := |\mathbf{t}_i|$ ($i < \omega$).

XVII. **Definition.** Let μ be a weak homogeneity system on X and let θ be an ordinal. The tree $\text{mstw}(\mu, \theta)$ on $X \times \text{Ord}$ consists of pairs (s, u) such that

$$u_0 < \theta \wedge (\forall i, j < |u|)(\mathbf{t}_i \triangleleft \mathbf{t}_j \implies j_{(s \upharpoonright l_i, \mathbf{t}_i), (s \upharpoonright l_j, \mathbf{t}_j)}^\mu(u_i) > u_j).$$

XVIII. **Remark.** For some $x \in X^\omega$, the tree $\text{mstw}(\mu, \theta)_x$ searches for a witness to the fact that all $\text{Ult}(V, \mu_{x,y})$ are ill-founded ($y \in \mathbb{R}$).

XIX. **Lemma.** For $\theta \geq (|Z|^\mathbb{C})^+$, we have $p[\text{mstw}(\mu, \theta)] = W_\mu^\mathbb{C}$.

Proof.

- 1° Inclusion ($\subseteq$) is obvious. Let us consider inclusion ($\supseteq$).
- 2° Suppose that for some $x \in X^\omega$, we have that every $\text{Ult}(V, \mu_{x,y})$ is ill-founded ($y \in \mathbb{R}$).
- 3° For $y \in \mathbb{R}$, there are $\alpha_n^y \in \text{Ord}$ ($n < \omega$) such that

$$j_{(x \upharpoonright n, y \upharpoonright n), (x \upharpoonright (n+1), y \upharpoonright (n+1))}^\mu(\alpha_n^y) > \alpha_{n+1}^y.$$

These ordinals can be chosen so that $\alpha_0^y < |Z|^+$ for every y .

- 4° For $t \in \omega^{<\omega}$, let $\hat{t} := \{y \in \mathbb{R} : t \triangleleft y\}$.
- 5° Let F be the class consisting of functions $\rho : \hat{t} \rightarrow \text{Ord}$ ($t \in \omega^{<\omega}$). Let it be ordered by relation $<_F$ defined by

$$\rho <_F \sigma : \iff t \supseteq u \wedge (\forall y \triangleright t)(\rho(y) < \sigma(y)),$$

where $\rho : \hat{t} \rightarrow \text{Ord}$ and $\sigma : \hat{u} \rightarrow \text{Ord}$.

- 6° $<_F$ is well-founded. Let rank_F denote its rank.
- 7° Classes $F, <_F, \text{rank}_F$ are absolute between all the models $\text{Ult}(V, \mu_{s,t})$, where $s \triangleleft x$. This is due to the fact that all these models are $\mathfrak{c}$ -closed.
- 8° For $t \in \omega^{<\omega}$, let $\rho_t : \hat{t} \rightarrow \text{Ord}$ be defined by setting $\rho_t(y) := \alpha_{|t|}^y$ ($y \triangleright t$). Obviously, $\rho_t \in F$.
- 9° For $t \in \omega^{<\omega}$, let

$$\beta_t := \text{rank}_F(\rho_t).$$

Note that $|\beta_0| \leq |Z|^\mathfrak{c}$.

- 10° For $t \triangleleft u$, we have

$$\begin{aligned} j_{(x \upharpoonright |t|, t), (x \upharpoonright |u|, u)}(\beta_t) > \beta_u &\iff \text{rank}_F(j_{(x \upharpoonright |t|, t), (x \upharpoonright |u|, u)}(\rho_t)) > \text{rank}_F(\rho_u) \\ &\iff j_{(x \upharpoonright |t|, t), (x \upharpoonright |u|, u)}(\rho_t) >_F \rho_u \\ &\iff (\forall y \triangleright u)(j_{(x \upharpoonright |t|, t), (x \upharpoonright |u|, u)}(\rho_t)(y) > \rho_u(y)) \\ &\iff (\forall y \triangleright u)(j_{(x \upharpoonright |t|, t), (x \upharpoonright |u|, u)}(\alpha_{|t|}^y) > \alpha_{|u|}^y) \\ &\iff \top. \end{aligned}$$

- 11° Thus, taking $z := (\beta_{t_i} : i < \omega)$, we have $(x, z) \in [\text{mstw}(\mu, \theta)]$ and consequently, $x \in p[\text{mstw}(\mu, \theta)]$.

□

XX. Proposition. Suppose that:

- $|Z| \geq \kappa$
- T is a tree on $X \times Z$ in V ;
- μ is a weak κ -homogeneity system for T ;
- $\theta \geq (|Z|^\mathfrak{c})^+$
- $U := \text{mstw}(\mu, \theta)$.

Then:

- a. U is κ -absolute complement for T ;
- b. if G be $<\kappa$ -generic over V , then μ^G is a weak κ -homogeneity system for T in $V[G]$.

Proof.

- 1° Let G be arbitrary $<\kappa$ -generic over V . We want to show that in $V[G]$:
 - a. $p[U] = p[T]^\mathfrak{G}$;
 - b. $p[T] = \{x \in X^\omega : (\exists y \in \mathbb{R})(\text{Ult}(V[G], \mu_{x,y}^G) \text{ is well-founded})\}$.
- 2° ($\subseteq$) in **a** is obvious since $V \models p[T] \cap p[U] = \emptyset$.
- 3° ($\supseteq$) of **b**:

- 1' Suppose that $\mu_{x,y}^G$ is complete for some $y \in \mathbb{R}$.
- 2' We have $\mu_{s,y||s|}^G(T_s) = \mu_{s,y||s|}(T_s) = 1$ for every $s \triangleleft x$.
- 3' It follows that there is a fiber z through $(T_s : s \triangleleft x)$.
- 4' z witnesses $x \in p[T]$.
- 4° ($\supseteq$) of **a**:
 - 1' Suppose $x \in p[U]^\mathbb{G}$.
 - 2' We have $U = \text{mstw}(\mu^G, \theta)$.
 - 3' By 1.3-XIX, we have that $\text{Ult}(V[G], \mu_{x,y}^G)$ is well-founded for some $y \in \omega^\omega$.
 - 4' Thus, $x \in p[T]$ by 3°.
- 5° We have established **a**.
- 6° ($\subseteq$) of **b**:
 - 1' Let $x \in p[T]$.
 - 2' By **a**, we have $x \in p[U]^\mathbb{G}$.
 - 3' Since $U = \text{mstw}(\mu^G, \theta)$, we have that $\text{Ult}(V[G], \mu_{x,y}^G)$ is well-founded for some $y \in \mathbb{R}$, by 1.3-XIX.
- 7° This finishes the proof.

□

XXI. **Proposition.** If $A \subseteq X^\omega$ is weakly κ -homogeneous, then it is κ -universally Baire.

Proof. Immediately from the previous proposition. □

XXII. **Proposition.** Let κ be an inaccessible cardinal and let $X \in V_\kappa$. Suppose that $A \subseteq X^\omega$ is κ -universally Baire. Then there are κ -absolutely complementing trees T, U on $X \times \kappa$ with $p[T] = A$.

Proof.

- 1° Let T^* on $X \times Z$ and U^* on $X \times W$ be κ -absolutely complementing trees such that $p[T] = A$.
- 2° Let η be large enough regular.
- 3° Let T be the subset of T^* consisting of elements definable over

$$\mathcal{H} := (H_\eta, \in, T^*, U^*, \kappa, R, x : x \in V_\kappa),$$

where R is some well-order on H_η .

- 4° **Claim.** T is a subtree of T^* of cardinality at most κ and $p[T] = p[T^*]$.

Proof.

- 1' If $s \in T$ and $s' \triangleleft s$, then $s' = s \upharpoonright |s'|$ is definable from s in $\mathcal{H}$ and consequently, $s' \in T$.
- 2' Thus, T is a subtree of T^* and $p[T] \subseteq p[T^*]$.
- 3' For the other inclusion, suppose that $x \in p[T^*]$.
- 4' $\mathcal{H} \models \exists z((x, z) \in [T^*])$. Let z_0 be the R -minimal witness to this.
- 5' Since $x \in X^\omega \subseteq V_\kappa$ and T^* are named in $\mathcal{H}$, we conclude that z_0 is definable over $\mathcal{H}$.
- 6' Then every $u_n := z_0 \upharpoonright n \in T^*$ is definable over $\mathcal{H}$ and consequently, $(x \upharpoonright n, u_n) \in T$.
- 7' Thus, $(x, z_0) \in [T]$, witnessing $x \in p[T]$.
- 8' The cardinality bound is obvious.

□

- 5° U is defined analogously with respect to U^* , it is a subtree of U^* of cardinality at most κ , and $p[U] = p[U^*]$.
- 6° **Claim.** For every $G \rightsquigarrow_{<\kappa} V$,

$$V[G] \models p[T] \sqcup p[U] = X^\omega.$$

Proof.

- 1' Let $\mathbb{P} \in V_\kappa$ be the poset for G and let us work in $V[G]$. The only thing to verify is that $X^\omega \subseteq p[T] \cup p[U]$. Let $x \in X^\omega$ be arbitrary.
- 2' In V , (T^*, U^*) are κ -absolutely complementing, so in $V[G]$, $x \in p[T^*] \cup p[U^*]$. We may assume w.l.o.g. that $x \in p[T^*]$. We want to show that $x \in p[T]$.
- 3' There is a name $\dot{x} \in V \upharpoonright \kappa$ for x .
- 4' Since $H_\eta[G] \models x \in p[T^*]$, there is $a \in \mathbb{P}$ such that

$$H_\eta \models (a \Vdash_{\mathbb{P}} \dot{x} \in p[T^*]).$$

- 5' $H_\eta \models \exists \dot{z}(a \Vdash_{\mathbb{P}} (\dot{x}, \dot{z}) \in [T^*])$
- 6' Taking $\dot{z}_0$ to be the R -minimal witness, we conclude that $\dot{z}_0$ is definable over $\mathcal{H}$ (note that $\mathbb{P}$ and a are named in $\mathcal{H}$).
- 7' We are done if we show that $(x, \dot{z}_0^G)$ is a branch through T .
- 8' Let $n < \omega$, let $s := x \upharpoonright n$, and let $u := \dot{z}_0^G \upharpoonright n$. We need to verify that $(s, u) \in T$.
- 9' There is $b \in G \subseteq V \upharpoonright \kappa$ with $b \leq a$ and $b \Vdash \dot{z}_0 \upharpoonright n = \check{u}$.
- 10' Since b is named in $\mathcal{H}$, we have that u is definable in $\mathcal{H}$. Consequently, $(s, u) \in T$.

□

- 7° Thus, T, U are as required.

□

XXIII. **Lemma.** Let λ be a cardinal, let $C \subseteq \lambda$ be a cofinal set of cardinals, and for every $\kappa \in C$, let T_κ be a κ -absolutely complemented tree on $X \times Z_\kappa$ such that $p[T_\kappa] = A$. Let T be the tree consisting of tuples (s, u, k) such that for some $n < \omega$ and for some $\kappa \in C$: $s \in X^n$, $u \in Z_\kappa^n$, $(s, u) \in T_\kappa$, $k \in \{\kappa\}^n$. Then T is λ -absolutely complemented and $p[T] = A$.

Proof.

- 1° A branch through T yields, after disregarding the last coordinate with constant value κ , a branch through T_κ . A branch through T_κ yields, after adding the last coordinate with constant value κ , a branch through T .
- 2° Thus, $p[T] = A$.
- 3° For every $\kappa \in C$, let us fix tree U_κ on $X \times W_\kappa$ that is κ -absolute complement of T_κ .
- 4° Let U be the tree consisting of tuples (s, u, k) such that for some $n < \omega$ and for some $\kappa \in C$: $s \in X^n$, $u \in W_\kappa^n$, $(s, u) \in U_\kappa$, $k \in \{\kappa\}^n$.
- 5° We want to show that (T, U) are λ -absolutely complemented.
- 6° A branch through U yields, after disregarding the last coordinate with constant value κ , a branch through U_κ . A branch through U_κ yields, after adding the last coordinate with constant value κ , a branch through U .
- 7° Thus, $p[T] \cap p[U] = \emptyset$.
- 8° It remains to show that, for any $g \rightsquigarrow_{<\lambda} V$, we have $p[T] \cup p[U] = (X^\omega)^V[g]$.
- 9° $g \rightsquigarrow_{<\kappa} V$ for some $\kappa \in C$.
- 10° We work in $V[g]$.
- 11° Let $x \in X^\omega$ be arbitrary. We need to show $x \in p[T] \cup p[U]$.
- 12° We have $x \in p[T_\kappa]$ or $x \in p[U_\kappa]$. Let us assume w.l.o.g. that $x \in p[T_\kappa]$.
- 13° This is witnessed by a branch (x, z) through T_κ .
- 14° $(x, z, (\kappa : n < \omega))$ is then a branch through T witnessing $x \in p[T]$.

□

XXIV. **Proposition.** Let λ be a cardinal and let $A \subseteq X^\omega$ be κ -universally Baire for cofinally many $\kappa < \lambda$. Then A is λ -universally Baire.

Proof. Immediately from the previous lemma. □

XXV. A *blueprint for a continuous function* $X^\omega \rightarrow Y^\omega$ is a function $f : X^{<\omega} \rightarrow Y^{<\omega}$ that has the following properties for $s, t \in X^{<\omega}$:

- a. $s \triangleleft t \implies f(s) \trianglelefteq f(t)$
- b. $|f(s)| - |f(s \upharpoonright (|s| - 1))| \leq 1$
- c. $\forall s \forall n \exists t (s \triangleleft t \wedge |f(t)| \geq n)$.

Such a blueprint determines a continuous function $\bar{f} : X^\omega \rightarrow Y^\omega$ in the following manner:

$$\bar{f}(x) := \bigcup_{n < \omega} f(x \upharpoonright n).$$

It is easily seen that for every continuous $F : X^\omega \rightarrow Y^\omega$, there is a blueprint f such that $F = \bar{f}$.

XXVI. Proposition.

- a. If $A \subseteq X^\omega$ is closed, then it is κ -universally Baire for every κ . Furthermore, if T is a tree on X such that $A = [T]$ and if $g \rightsquigarrow V$, then $A^g = [T]^{V[g]}$.
- b. If $f : X^\omega \rightarrow Y^\omega$ is continuous, then it is κ -universally Baire for every κ . Furthermore, for every $g \rightsquigarrow V$, we have $f^g : X^\omega \rightarrow Y^\omega$ and it is continuous; if h is a blueprint for a continuous function such that $f = \bar{h}^V$, then $f^g = \bar{h}^{V[g]}$.
- c. If $A \subseteq X^\omega$ is κ -universally Baire, then so is $A^\mathbb{C}$. Furthermore, for every $g \rightsquigarrow_{<\kappa} V$, we have $(A^\mathbb{C})^g = (A^g)^\mathbb{C}$.
- d. If $A, B \subseteq X^\omega$ are κ -universally Baire, then so are $A \cup B$ and $A \cap B$. Furthermore, for every $g \rightsquigarrow_{<\kappa} V$, we have $(A \cup B)^g = A^g \cup B^g$ and $(A \cap B)^g = A^g \cap B^g$.
- e. If $B \subseteq Y^\omega$ is κ -universally Baire and $f : X^\omega \rightarrow Y^\omega$ is continuous, then $f^{-1}[B]$ is also κ -universally Baire and for every $g \rightsquigarrow_{<\kappa} V$, we have

$$(f^{-1}[B])^g = (f^g)^{-1}[B^g].$$

Proof.

Part a.

- 1° Let T be a tree on X such that $A = [T]$.
- 2° Let $S := \{(s, s) : s \in T\}$. It is clear that $p[S] = [T]$ in every generic extension of V .
- 3° Let U be the tree on $X \times X^{<\omega}$ consisting of pairs $(s, (t)_{i < n})$, where $s \in X^n$, $t \notin T$, and s is comparable with t .
- 4° It is clear that $p[U] = [T]^\mathbb{C}$ in every generic extension of V .
- 5° The conclusion is now obvious.

Part b.

- 6° $f \subseteq X^\omega \times Y^\omega$ is closed, so it is κ -universally Baire for every κ .
- 7° We only need to show that for a blueprint h for f and $g \rightsquigarrow V$, we have $f^g = \bar{h}^{V[g]}$.
- 8° Let $T := \{(s, t) : s \in X^{<\omega}, t \supseteq h(s), |t| = |s|\}$. We see that T is a tree on $X \times Y$ and $[T] = \bar{h}$ in every generic extension of V . By [a](#), this suffices for the conclusion.

Part c.

9° Let T, U be κ -absolutely complemented such that $p[T] = A$. Then pair (U, T) witnesses that $A^{\mathfrak{C}}$ is κ -universally Baire.

10° Also, $(A^{\mathfrak{C}})^g = p[U]^{V[g]} = (p[T]^{\mathfrak{C}})^{V[g]} = (A^g)^{\mathfrak{C}}$.

Part d.

11° It suffices to verify the case of the union.

12° Let T_A, U_A and T_B, U_B be κ -absolutely complemented such that $p[T_A] = A$ and $p[T_B] = B$.

13° $T := \{(s, u, c) : s \in X^{<\omega} \wedge ((u \in (T_A)_s \wedge c \in \{0\}^{|s|}) \vee (u \in (T_B)_s \wedge c \in \{1\}^{|s|}))\}$

14° $U := \{(s, u, v) : s \in X^{<\omega} \wedge u \in (U_A)_s \wedge v \in (U_B)_s\}$

15° T, U are κ -absolutely complemented and $p[T] = A \cup B$.

16° Also, $(A \cup B)^g = p[T]^{V[g]} = (p[T_A] \cup p[T_B])^{V[g]} = A^g \cup B^g$.

Part e.

17° Let T', U' be trees on $Y \times Z$ and $Y \times W$ that are κ -absolutely complemented and such that $p[T'] = B$. We may assume that $0 \notin Z \cup W$. Let h be a blueprint for f .

18° We describe tree T on $X \times (Z \cup \{0\})$ by describing T_s recursively on the length of a node $s \in X^{<\omega}$.

1' Case $|s| = 0$ is clear.

2' Let us consider recursively case $|s| = n + 1$.

3' If $h(s) = h(s \restriction n)$, then T_s is built in the following manner:

- a. take any $u \in T_{s \restriction n}$;
- b. put $u \frown 0$ in T_s .

4' If $h(s) \triangleright h(s \restriction n)$, then T_s is built in the following manner:

- a. take any $u \in T_{s \restriction n}$;
- b. omit all the zeroes to get $u' \in T'_{h(s \restriction n)}$;
- c. take any $u'' \triangleright u'$ in $T'_{h(s)}$;
- d. put $u \frown u''(|u''| - 1)$ in T_s .

19° Tree U on $Y \times (W \cup \{0\})$ is constructed analogously from tree U' .

20° $p[T] = \bar{h}^{-1}[p[T']]$ and $p[U] = \bar{h}^{-1}[p[U']]$ in every generic extension of V , thus concluding the argument.

□

1.4 Transfer Theorems

- I. We have so far established that every κ -homogeneous set is weakly κ -homogeneous and that every weakly κ -homogeneous set is κ -universally Baire. In this section, we aim to reverse (in some sense) these implications. Theorems establishing such reversals are often referred to as *transfer theorems*.

II. **Martin-Steel Transfer Theorem.** Let δ be Woodin and let $X \in V_\delta$. Suppose that $A \subseteq X^\omega$ is weakly δ^+ -homogeneous. Then $A^\mathbb{C}$ is $<\delta$ -homogeneous (i.e. it is κ -homogeneous for every $\kappa < \delta$).

Proof. This was the main theorem of [Kasum, 2021a] (theorem 9.2). We omit its proof here. $\square$

III. The other transfer we are interested in, which will be proven below, is *Woodin's Transfer Theorem*. We start with some lemmas.

IV. **Lemma.** If δ is Woodin and $G \rightsquigarrow V^{\mathbb{Q}_\delta}$, then for every $x \in V_\delta^\omega \cap V[G]$, there is a name $\dot{x} \in V^{\mathbb{Q}_\delta} \cap V_\delta$ such that $\dot{x}^G = x$.

Proof.

1° Let τ be a name for x . It suffices to show that

$$D := \{a \in \mathbb{Q}_\delta : (\exists \dot{x} \in V \upharpoonright \delta)(a \Vdash \dot{x} = \tau)\}$$

is dense.

2° Let A_n be a maximal anti-chain whose elements decide $\tau(n)$ ($n < \omega$).

3° Let $a_0 \in \mathbb{Q}_\delta$ be arbitrary. We want to find $a \in D$ such that $a \leq a_0$.

4° There is an inaccessible $\kappa < \delta$ such that $A_n \cap \mathbb{Q}_\kappa$ is semi-proper for every $n < \omega$ and $a_0 \in V_\kappa$.

5° Let a consist of all countable $M \prec (H(\kappa^+), \in, (A_n \cap \mathbb{Q}_\kappa : n < \omega), a_0)$ that capture every $A_n \cap \mathbb{Q}_\kappa$ and such that $M \cap (\cup a_0) \in a_0$.

6° **Claim.** $a \in \mathbb{Q}_\delta$

Proof.

1' Let F be an algebra on $H(\kappa^+)$. We want to find $M \in a$ closed for F .

2' Let $M_0 \prec (H_\delta, \in, \kappa, F, (A_n \cap \mathbb{Q}_\kappa : n < \omega), a_0)$ be countable such that $M \cap (\cup a_0) \in a_0$.

3' Suppose that $M_n \prec (H_\delta, \in, \kappa, (A_n \cap \mathbb{Q}_\kappa : n < \omega))$ has been defined.

4' Since $A_n \cap \mathbb{Q}_\kappa$ is semi-proper, we can choose $M_{n+1} \prec H_\delta$ such that:

- a. $M_n \subseteq M_{n+1}$
- b. $M_{n+1} \cap V_\kappa$ end-extends $M_n \cap V_\kappa$;
- c. M_{n+1} captures $A_n \cap \mathbb{Q}_\kappa$.

5' Let $N := \bigcup_{n < \omega} M_n \prec H_\delta$.

6' It is clear that $M := N \cap H(\kappa^+) \in a$.

7' Since $F \in N$, model M is closed for F , so it is as required. $\square$

7° Obviously, $a \leq a_0$.

8° **Claim.** $|\{b \in A_n : b \parallel a\}| < \delta$ for every $n < \omega$.

Proof.

- 1' Let $b \in A_n$ be arbitrary such that $b \parallel a$.
- 2' Let $c \leq a, b$ and let $M \prec (H_\delta, \in, b)$ be such that $M \cap (\cup c) \in c$.
- 3' Then $M \cap H(\kappa^+) \in a$, so there is $b' \in M \cap A_n \cap \mathbb{Q}_\kappa$ such that $M \cap (\cup b') \in b'$.
- 4' **Subclaim.** $b = b'$

Proof.

- 1'' Assume otherwise.
- 2'' Then $b \perp b'$, as witnessed by some algebra F .
- 3'' $b, b' \in M$, so F can be chosen in M .
- 4'' Then M is closed for F , while $M \cap (\cup b) \in b$ and $M \cap (\cup b') \in b'$. This is a contradiction.

□

5' Thus, $b = b' \in M \cap V_\kappa \subseteq V_\kappa$.

6' We have shown $A_n \subseteq V_\kappa$.

□

9° **Claim.** $a \in D$

Proof.

1' Let

$$\dot{x} := \{((n, A), b) : n < \omega, A \in V_\delta, b \in A_n, b \parallel a, b \Vdash \tau(n) = A\}.$$

2' $\dot{x} \in V^{\mathbb{Q}_\delta} \cap V_\delta$ and $a \Vdash \tau = \dot{x}$.

□

□

V. **Lemma.** Let δ be a Woodin cardinal and let $X \in V_\delta$. Suppose that T, U are δ^+ -absolutely complemented trees on $X \times Z$ and $X \times W$. Then there are subtrees $T^* \subseteq T$ and $U^* \subseteq U$ such that

- a. $|T^*|, |U^*| \leq \delta$
- b. $p[T^*] = p[T]$ and $p[U^*] = p[U]$;
- c. $V^{\mathbb{Q}_\delta} \models p[T^*] = p[T] \wedge p[U^*] = p[U]$.

Proof.

1° Let η be large enough regular.

2° Let T^* be the subset of T consisting of elements definable over

$$\mathcal{H} := (H_\eta, \in, T, U, \delta, R, x : x \in V_\delta),$$

where R is some well-ordering of H_η .

3° Obviously $|T^*| \leq \delta$.

4° **Claim.** T^* is a subtree of T and $p[T^*] = p[T]$.

Proof.

- 1' If $s \in T^*$ and $s' \triangleleft s$, then $s' = s \upharpoonright |s'|$ is definable from s in $\mathcal{H}$ and consequently, $s' \in T^*$.
- 2' Thus, T^* is a subtree of T and $p[T^*] \subseteq p[T]$.
- 3' For the other inclusion, suppose that $x \in p[T]$.
- 4' $\mathcal{H} \models \exists z((x, z) \in [T])$. Let z_0 be the R -minimal witness to this.
- 5' Since $x \in X^\omega \subseteq V_\delta$ and T are named in $\mathcal{H}$, we conclude that z_0 is definable over $\mathcal{H}$.
- 6' Then every $u_n := z_0 \upharpoonright n \in T$ is definable over $\mathcal{H}$ and consequently, $(x \upharpoonright n, u_n) \in T^*$.
- 7' Thus, $(x, z_0) \in [T^*]$, witnessing $x \in p[T^*]$.

□

5° U^* is defined analogously with respect to U , it is a subtree of U of cardinality at most δ , and $p[U^*] = p[U]$.

6° **Claim.** For every $G \rightsquigarrow V^{\mathbb{Q}_\delta}$, $p[T^*]^{V[G]} = p[T]^{V[G]}$ and $p[U^*]^{V[G]} = p[U]^{V[G]}$.

Proof.

- 1' Let us verify the case of T . The case of U is analogous.
- 2' Obviously $p[T^*] \subseteq p[T]$.
- 3' Let $x \in p[T]$ be arbitrary. We want to show that $x \in p[T^*]$.
- 4' There is a name $\dot{x} \in V_\delta$ for x (cf. 1.4-IV).
- 5' Since $H_\eta[G] \models x \in p[T]$, there is $a \in \mathbb{Q}_\delta$ such that
$$H_\eta \models (a \Vdash_{\mathbb{Q}_\delta} \dot{x} \in p[T]).$$
- 6' $H_\eta \models \exists \dot{z}(a \Vdash_{\mathbb{Q}_\delta} (\dot{x}, \dot{z}) \in [T])$
- 7' Taking $\dot{z}_0$ to be the R -minimal witness, we conclude that $\dot{z}_0$ is definable over $\mathcal{H}$ (note that $\dot{x}, a$ are named in $\mathcal{H}$, while $\mathbb{Q}_\delta$ is definable there).
- 8' We are done if we show that $(x, \dot{z}_0^G)$ is a branch through T^* .
- 9' Let $n < \omega$, let $s := x \upharpoonright n$, and let $u := \dot{z}_0^G \upharpoonright n$. We need to verify that $(s, u) \in T^*$.
- 10' There is $b \in G \subseteq V_\delta$ with $b \leq a$ and $b \Vdash \dot{z}_0 \upharpoonright n = \check{u}$.
- 11' Since b is named in $\mathcal{H}$, we have that u is definable in $\mathcal{H}$. Consequently, $(s, u) \in T^*$.

□

7° Thus, T^*, U^* are as required.

□

VI. Theorem. (Woodin) Let δ be Woodin and let $X \in H_{\omega_1}$. Suppose that tree T on $X \times Z$ is δ^+ -absolutely complemented. Then T is weakly κ -homogeneous for every $\kappa < \delta$.

Proof.

- 1° We may assume that T is pruned.
- 2° By the previous lemma, there are (pruned) subtree $T^* \subseteq T$ and tree U^* on $X \times \delta$ such that:
 - a. $|T^*| \leq \delta$
 - b. $p[T^*] = p[T] = X^\omega - p[U^*]$
 - c. $V^{\mathbb{Q}_\delta} \models p[T^*] = p[T] = X^\omega - p[U^*]$.
- 3° We may assume that T^* is also a tree on $X \times \delta$.
- 4° Let us show that T^* is weakly κ -homogeneous, for every $\kappa < \delta$.
- 5° We may assume w.l.o.g. that κ is T^* -reflecting up to δ , i.e. for every inaccessible $\lambda \in (\kappa, \delta)$, there is embedding $j_\lambda : V \prec M_\lambda$ such that:
 - a. $\text{crit } j_\lambda = \kappa$
 - b. $V_\lambda \subseteq M_\lambda$
 - c. $j_\lambda(T) \cap V_\lambda = T \cap V_\lambda$.
- 6° For $\lambda \in (\kappa, \delta)$ inaccessible and $(s, u) \in T^* \cap V_\lambda$, we define measure $\Sigma_\lambda(s, u)$ on $\kappa^{|s|}$ as the measure derived from j_λ at $u \in \lambda^{|s|}$.
- 7° For every inaccessible $\lambda \in (\kappa, \delta)$, function Σ_λ has the following properties.
 - a. If $(s, u) \triangleleft (t, v)$, then $\Sigma_\lambda(t, v)$ projects to $\Sigma_\lambda(s, u)$.
 - b. For every $(x, z) \in [T^* \cap V_\lambda]$, tower

$$(\Sigma_\lambda(x \restriction n, z \restriction n) : n < \omega)$$

is complete.

(7°b follows from the fact that the ultrapowers by measures $\Sigma_\lambda(x \restriction n, z \restriction n)$ are j_λ -realizable in a commutative manner.)

- 8° Let $G \rightsquigarrow V^{\mathbb{Q}_\delta}$ and let $k := j_G : V \prec \text{Ult}(V, G) =: \mathfrak{M}$. We want to show that $k(T^*)$ is weakly $k(\kappa)$ -homogeneous in $\mathfrak{M}$.
- 9° Let σ be the set of all $k(\mu)$, where μ is a κ -complete measure on some κ^n ($n < \omega$) in V .
- 10° σ is the pointwise image by k of a set in V_δ . Thus, it belongs to and is countable in $\mathfrak{M}$.

11° **Claim.** In $\mathfrak{M}$, for every $x \in p[k(T^*)]$, there is a complete tower of measures $\mu \in \sigma^\omega$ such that for every $s \triangleleft x$, we have $\mu_{|s|}(k(T^*)_s) = 1$.

Proof.

- 1' Let $x \in p[k(T^*)]^\mathfrak{M}$ be arbitrary.
- 2' $x \in k(p[T^*]^V)$
- 3' Since $x \in X^\omega$, we have $x \in p[T^*]$ or $x \in p[U^*]$ (in $V[G]$).
- 4' **Subclaim.** $x \notin p[U^*]^{V[G]}$

Proof.

- 1'' Assume otherwise, as witnessed by some w .
- 2'' Hence, $(x \restriction n, w \restriction n) \in U^*$ for every $n < \omega$.
- 3'' This means $(x \restriction n, k(w \restriction n)) \in k(U^*)$ for every $n < \omega$.
- 4'' Since $\mathfrak{M}^\omega \subseteq \mathfrak{M}$, we have $w' := \bigcup_{n < \omega} k(w \restriction n) \in \mathfrak{M}$.
- 5'' Then $(x, w') \in [k(U^*)]^\mathfrak{M}$ and consequently, $x \in p[k(U^*)]^\mathfrak{M} = k(p[U^*]^V) = k((X^\omega)^V - p[T^*]^V)$, which contradicts 11°2'.

□

- 5' Hence, $x \in p[T^*]^{V[G]}$.
- 6' Let z be such that $(x, z) \in [T^*]^{V[G]}$.
- 7' There is V -inaccessible $\lambda \in (\kappa, \delta)$ such that $(x, z) \in [T^* \cap V_\lambda]^{V[G]}$.
- 8' Let $\mu := (k(\Sigma_\lambda)(x \restriction n, k(z \restriction n)) : n < \omega)$.
- 9' $k(\Sigma_\lambda)(x \restriction n, k(z \restriction n)) = k(\Sigma_\lambda)(x \restriction n, z \restriction n) \in \sigma$ for every $n < \omega$.
- 10' Since $\mathfrak{M}^\omega \subseteq \mathfrak{M}$, we $\mu \in \sigma^\omega$.
- 11' By 7°a and elementarity of k , we have that μ is a tower of measures.
- 12' For every $n < \omega$, we have $(x \restriction n, z \restriction n) \in T^* \cap V_\lambda$.
- 13' This means that for every $n < \omega$, we have $(x \restriction n, k(z \restriction n)) \in k(T^* \cap V_\lambda)$.
- 14' Since $\mathfrak{M}$ is closed for ω -sequences, we have

$$(x, \bigcup \{k(z \restriction n) : n < \omega\}) \in [k(T^* \cap V_\lambda)]^\mathfrak{M}.$$

- 15' By 7°b and elementarity of k , we conclude that μ is a complete tower in $\mathfrak{M}$.

- 16' For every $n < \omega$, we have

$$\mu_n(k(T^*)_{x \restriction n}) = k(\Sigma_\lambda)(x \restriction n, z \restriction n)(T^*_{x \restriction n}) = k(1) = 1.$$

(The penultimate equality is due to the fact that $(x \restriction n, z \restriction n) \in T^* \cap V_\lambda = j_\lambda(T^*) \cap V_\lambda \subseteq j_\lambda(T^*)$.)

□

12° Since every measure in σ can be seen as a measure on $k(\delta)^n$ (for the appropriate n), we conclude by 1.2-IX that $k(T^*)$ is $k(\kappa)$ -homogeneous in $\mathfrak{M}$.

13° Thus, T^* is weakly κ -homogeneous for every $\kappa < \delta$.

14° **Claim.** T is weakly κ -homogeneous for every $\kappa < \delta$.

Proof.

- 1' Let $\kappa < \delta$ be arbitrary.
- 2' Since T^* is weakly κ -homogeneous, 1.2-IX implies that there is countable σ such that:
 - a. for every $\mu \in \sigma$, there is $n < \omega$ such that μ is a κ -complete measure on δ^n ;
 - b. for every $x \in p[T^*]$, there exists complete tower $\mu \in \sigma^\omega$ such that $(\forall n < \omega)(\mu_n(T_{x \upharpoonright n}^*) = 1)$.
- 3' For every measure $\mu \in \sigma$ on δ^n , let $\tilde{\mu}$ denote the measure determined by

$$(\forall C \subseteq Z^n)(\tilde{\mu}(C) = \mu(C \cap \delta)).$$

- 4' Let $\sigma' := \{\tilde{\mu} : \mu \in \sigma\}$.

5' **Subclaim.**

- a. $|\sigma'| = \omega$
- b. for every $\tilde{\mu} \in \sigma'$, there is $n < \omega$ such that $\tilde{\mu}$ is a κ -complete measure on Z^n ;
- c. for every $x \in p[T]$, there exists complete tower $\tilde{\mu} \in \sigma'^\omega$ such that $(\forall n < \omega)(\tilde{\mu}_n(T_{x \upharpoonright n}) = 1)$.

Proof.

- 1'' Only the last point requires an explanation.
- 2'' Let $x \in p[T]$.
- 3'' Then $x \in p[T^*]$.
- 4'' There is a complete tower $\mu \in \sigma^\omega$ such that

$$(\forall n < \omega)(\mu_n(T_{x \upharpoonright n}^*) = 1).$$

- 5'' Let $\tilde{\mu} := (\tilde{\mu}_n : n < \omega) \in \sigma'^\omega$.

- 6'' Clearly, $\tilde{\mu}$ is a complete tower.

- 7'' For $n < \omega$,

$$\tilde{\mu}_n(T_{x \upharpoonright n}) = \mu_n(T_{x \upharpoonright n} \cap \delta) \geq \mu_n(T_{x \upharpoonright n}^*) = 1.$$

□

- 6' By 1.2-IX, we conclude that T is weakly κ -homogeneous.

□

□

VII. **Woodin's Transfer Theorem.** Let δ be Woodin and let $X \in H_{\omega_1}$. Suppose that $A \subseteq X^\omega$ is δ^+ -universally Baire. Then A is weakly $<\delta$ -homogeneous (i.e. it is weakly κ -homogeneous for every $\kappa < \delta$).

Proof. Immediately from the previous theorem. $\square$

VIII. **Definition.** Let λ be a cardinal.

- a. $\text{Hom}_{<\lambda}$ is the collection of all $<\lambda$ -homogeneous sets $A \subseteq X^\omega$, where $X \in H_{\omega_1}$.
- b. $\text{WHom}_{<\lambda}$ is the collection of all weakly $<\lambda$ -homogeneous sets $A \subseteq X^\omega$, where $X \in H_{\omega_1}$.
- c. UB_λ is the collection of all λ -universally Baire sets $A \subseteq X^\omega$, where $X \in H_{\omega_1}$.

IX. **Theorem.** Let λ be a limit of Woodin cardinals. Then

$$\text{Hom}_{<\lambda} = \text{WHom}_{<\lambda} = \text{UB}_\lambda.$$

This collection of sets is closed for projections, unions, intersections, complements, and it is downward closed for Wadge reduction. Every set in this collection is determined.

Proof.

- 1° By Woodin's Transfer Theorem, we immediately have $\text{WHom}_{<\lambda} = \text{UB}_{<\lambda}$.
- 2° By 1.3-XXVI and the previous point, if $A \in \text{WHom}_{<\lambda}$, then also $A^c \in \text{WHom}_{<\lambda}$.
- 3° By Martin-Steel Transfer Theorem, we have $A \in \text{Hom}_{<\lambda}$.
- 4° We conclude that $\text{WHom}_{<\lambda} = \text{Hom}_{<\lambda}$.
- 5° The closure for projections follows from 1.2-XI.
- 6° The closure for unions, intersections, complements, and Wadge reduction follows from 1.3-XXVI.
- 7° Determinacy follows from 1.1-XVII.

$\square$

X. **Notation.** By *the second-order arithmetic*, we shall mean structure

$$\mathbb{R} := (\omega, \omega^\omega; \text{app}, +, \cdot, \text{pow}, 0, 1, \leq),$$

where

$$\text{app} : \omega^\omega \times \omega \rightarrow \omega : (x, n) \mapsto x(n), \quad \text{pow} : \omega \times \omega \rightarrow \omega : (m, n) \mapsto m^n.$$

XI. **Theorem.** Let $\bar{x}$ be an n -tuple and let $\phi(\bar{x}, \dot{A})$ be a formula in the second-order language of structure $\mathbb{R}$, with additional n_0 -ary relation symbol $\dot{A}$. Suppose that λ is a limit of Woodin cardinals and that $A \subseteq \mathbb{R}^{n_0}$ is λ -universally Baire. Then set

$$A_\phi := \{\bar{x} \in \mathbb{R}^n : (\mathbb{R}, A) \models \phi[\bar{x}, A]\}$$

is λ -universally Baire. Moreover, if $g \rightsquigarrow_{<\lambda} V$, then

$$A_\phi^g = \{\bar{x} \in (\mathbb{R}^g)^n : (\mathbb{R}^g, A^g) \models \phi[\bar{x}, A^g]\}.$$

Proof.

- 1° We may assume w.l.o.g. that ϕ does not contain first order variables.
- 2° The proof is by induction on complexity of ϕ .
- 3° Let us consider the base case of a formula of the form $R(t_0, t_1)$, where $R \in \{=, \leq\}$ and t_0, t_1 are terms.
- 4° By recursion on complexity of t_0, t_1 , we construct trees T and U on ω such that for every $g \rightsquigarrow V$, we have

$$\{x \in \mathbb{R} : t_0(x) = t_1(x)\} = [T], \quad \{x \in \mathbb{R} : t_0(x) \leq t_1(x)\} = [U]$$

in $V[g]$.

- 5° The conclusion now follows by part [a](#) of [1.3-XXVI](#).
- 6° Let us consider the base case of a formula of the form $\dot{A}(t_0 \dots t_{n_0-1})$, where t_i are terms.
- 7° By recursion on complexity of terms $t_0 \dots t_{n_0-1}$, we can define a blueprint $f : \omega^{<\omega} \rightarrow \omega^{<\omega}$ for a continuous function such that in every generic extension of V ,

$$\bar{f} : x \mapsto (t_0(x), \dots, t_{n_0-1}(x)).$$

- 8° By part [e](#) of [1.3-XXVI](#),

$$\{x \in \mathbb{R} : A(t_0(x), \dots, A_{n_0-1}(x))\} = \bar{f}^{-1}[A]$$

is λ -universally Baire and

$$\begin{aligned} \{x \in \mathbb{R} : A(t_0(x), \dots, A_{n_0-1}(x))\}^g &= (\bar{f}^{-1}[A])^g = \\ (\bar{f}^g)^{-1}[A^g] &= \{x \in \mathbb{R}^g : A^g(t_0(x), \dots, t_{n_0-1}(x))\}. \end{aligned}$$

- 9° The inductive cases of formulas of the form $\neg\psi$, $\psi \vee \theta$, and $\psi \wedge \theta$ follow from parts [c](#) and [d](#) of [1.3-XXVI](#).
- 10° Let us consider the inductive case of a formula of the form $\exists y\psi(x, y, \dot{A})$.
- 11° From the previous theorem, we have that

$$B := \{x \in \mathbb{R} : (\mathbb{R}, A) \models (\exists y\psi)[x, A]\} = p\{(x, y) : (\mathbb{R}, A) \models \psi[x, y, A]\}$$

is λ -universally Baire.

- 12° It remains to verify that for every $g \rightsquigarrow_{<\lambda} V$, we have

$$B^g = \{x \in \mathbb{R}^g : (\mathbb{R}^g, A^g) \models (\exists y\psi)[x, A^g]\}.$$

- 13° Let $\kappa < \lambda$ be such that $g \rightsquigarrow_{<\kappa} V$.
- 14° Let T be λ -absolutely complemented tree on $\omega \times \omega \times Z$ such that

$$p[T] = \{(x, y) : (\mathbb{R}, A) \models \psi[x, y, A]\}.$$

- 15° Tree $T' := \{(s, (t, u)) : (s, t, u) \in T\}$ projects to $pp[T]$ in every generic extension of V .
- 16° Let $\delta \in (\kappa, \lambda)$ be Woodin.
- 17° Since T is δ^+ -absolutely complemented, by Woodin's Transfer Theorem, it is weakly κ -homogeneous.
- 18° By 1.2-X, tree T' is weakly κ -homogeneous.
- 19° By 1.3-XX, tree T' is κ -absolutely complemented.
- 20° Since $p[T'] = pp[T] = B$, we have

$$\begin{aligned}
B^g &= p[T']^{V[g]} = pp[T]^{V[g]} = p^{V[g]}p[T]^{V[g]} = \\
&= p^{V[g]}\{(x, y) \in \mathbb{R}^2 : (\mathbb{R}, A) \models \psi[x, y, A]\}^g = \\
&= p^{V[g]}\{(x, y) \in (\mathbb{R}^g)^2 : (\mathbb{R}^g, A^g) \models \psi[x, y, A^g]\} = \\
&= \{x \in \mathbb{R}^g : (\mathbb{R}^g, A^g) \models (\exists y\psi)[x, A^g]\}.
\end{aligned}$$

(The penultimate equality follows from the inductive hypothesis.)

□

XII. Corollary. Suppose that λ is a limit of Woodin cardinals. Then every projective set of reals is λ -universally Baire. In particular, all projective sets are determined.

2 Tree Production Lemma

- I. In this section, we give a method of establishing that a set of reals is universally Baire.
- II. **Remark.** A *transitive hull* of V is a mapping $h : H \prec_n V$, with H transitive set and n large enough integer. We also write $h : H \prec V$. A *countable hull* of V is a transitive hull that is countable.
- III. **Definition.** Let $\phi(x, a)$ be a formula with parameter $a \in V$ and let κ be a cardinal.
 - a. Countable hull $h : \mathcal{H} \prec (V, \in, \kappa, a)$ is said to be *$<\kappa$ -generically correct* for $\phi(x, a)$ if for every g which is $<\kappa^{\mathcal{H}}$ -generic over H , we have
$$(\forall x \in \mathbb{R} \cap H[g])(H[g] \models \phi[x, a^{\mathcal{H}}] \iff V \models \phi[x, a]).$$
 - b. Transitive hull $h : \mathcal{H} \prec (V, \in, \kappa, a)$ is said to be *$<\kappa$ -generically correct* for $\phi(x, a)$ if the set of all countable $M \prec \mathcal{H}$ whose anti-collapse k satisfies that $h \circ k$ is $<\kappa$ -generically correct for $\phi(x, a)$ (in the sense of a), contains a club.
- IV. **Remark.** Clearly the two notions coincide in the case of a countable hull.

V. **Theorem.** Let $\phi(x, a)$ be a formula with parameter $a \in V$ and let κ be a regular uncountable cardinal. Let $h : \mathcal{H} \prec (V, \in, \kappa, a)$ be a transitive hull with $H_\kappa \subseteq \text{im } h$. Then the following are equivalent.

a. $A := \{x \in \mathbb{R} : \phi(x, a)\}$ is κ -universally Baire and

$$A^g = \{x \in \mathbb{R}^g : V[g] \models \phi[x, a]\}$$

for every $g \rightsquigarrow_{<\kappa} V$.

b. h is $<\kappa$ -generically correct for $\phi(x, a)$.

Proof.

Implication ($\Rightarrow$)

1° There are κ -absolutely complemented trees T, U such that for every $g \rightsquigarrow_{<\kappa} V$, we have

$$V[g] \models p[T] = \{x \in \mathbb{R} : \phi(x, a)\}.$$

2° By elementarity, they can be chosen in $\text{im } h$. Let $T^{\mathcal{H}} := h^{-1}(T)$ and $U^{\mathcal{H}} := h^{-1}(U)$.

3° Let $k : (M, \in, \bar{\kappa}, \bar{a}, \bar{T}, \bar{U}) \prec (\mathcal{H}, T^{\mathcal{H}}, U^{\mathcal{H}})$ be a countable hull. We want to show that $h \circ k$ is $<\kappa$ -generically correct for $\phi(x, a)$.

4° Let $g \rightsquigarrow_{<\bar{\kappa}} M$ be arbitrary. We want to show that for every $x \in \mathbb{R} \cap M[g]$, we have

$$M[g] \models \phi[x, \bar{a}] \iff V \models \phi[x, a].$$

5° In other words, we want to show

$$\bar{A}^g = A \cap M[g],$$

where $\bar{A} := k^{-1}(A) = \{x \in \mathbb{R} \cap M : M \models \phi[x, \bar{a}]\}$.

6° Suppose first that $x \in \bar{A}^g = p[\bar{T}]^{M[g]}$.

7° Let y be such that $(x \upharpoonright n, y \upharpoonright n) \in \bar{T}$ for every $n < \omega$.

8° Then $(x \upharpoonright n, (h \circ k)(y \upharpoonright n)) \in T$ for every $n < \omega$.

9° We conclude $x \in p[T] = A$.

10° Suppose now that $x \in (\mathbb{R} \cap M[g]) - \bar{A}^g$.

11° Then $x \in p[\bar{U}]^{M[g]}$.

12° The same reasoning as above now yields $x \in p[U] = A^{\mathbb{G}}$.

Implication ($\Leftarrow$)

13° Let $F : H^{<\omega} \rightarrow H$ be such that for every $M \in [H]^\omega$, if $F[M^{<\omega}] \subseteq M$, then $M \prec \mathcal{H}$ and its anti-collapse k satisfies that $h \circ k$ is $<\kappa$ -generically correct for $\phi(x, a)$.

14° We want to find trees T and U such that for every $g \rightsquigarrow_{<\kappa} V$, we have

$$V[g] \models p[T] = \{x \in \mathbb{R} : \phi(x, a)\} \wedge p[U] = \{x \in \mathbb{R} : \neg\phi(x, a)\}.$$

- 15° Let $\mathbf{t} : \omega \rightarrow \omega^{<\omega}$ be the standard bijection.
- 16° For $x \in \mathbb{R}$, we define T_x as the tree of attempts to construct a sequence $y \in H^\omega$ such that:
- a. $y(0) = \mathbb{P} \in H_\kappa$ is a poset;
 - b. $y(1) = \tau \in V^\mathbb{P} \cap H_\kappa$ is a name for a real;
 - c. $y(2) \in \mathbb{P}$ and $y(2) \Vdash \phi(\tau, a)$
 - d. $y(2n+3) = F(y \circ \mathbf{t}_n)$
 - e. $y(2n+4) \in \mathbb{P}$ and satisfies:
 - i. $y(2n+4) \leq y(2n+2)$;
 - ii. if $y(2n+3)$ is a dense subset of $\mathbb{P}$, then $y(2n+4) \in y(2n+3)$;
 - iii. $y(2n+4) \Vdash \tau \restriction n = x \restriction n$.
- 17° For $x \in \mathbb{R}$, we define U_x as the tree of attempts to construct a sequence $y \in H^\omega$ such that:
- a. $y(0) = \mathbb{P} \in H_\kappa$ is a poset;
 - b. $y(1) = \tau \in V^\mathbb{P} \cap H_\kappa$ is a name for a real;
 - c. $y(2) \in \mathbb{P}$ and $y(2) \Vdash \neg \phi(\tau, a)$
 - d. $y(2n+3) = F(y \circ \mathbf{t}_n)$
 - e. $y(2n+4) \in \mathbb{P}$ and satisfies:
 - i. $y(2n+4) \leq y(2n+2)$;
 - ii. if $y(2n+3)$ is a dense subset of $\mathbb{P}$, then $y(2n+4) \in y(2n+3)$;
 - iii. $y(2n+4) \Vdash \tau \restriction n = x \restriction n$.
- (The only difference is in 17°c.)
- 18° **Claim.** $p[T] \subseteq \{x \in \mathbb{R} : \phi(x, a)\}$
- Proof.*
- 1' Suppose $(x, y) \in T$ and let $M := \text{im } y$.
 - 2' M is closed for F , so $M \prec \mathcal{H}$. Let $k : (\overline{M}, \in, \overline{\kappa}, \overline{a}) \prec \mathcal{H}$ be the anti-collapse.
 - 3' $h \circ k$ is $<\kappa$ -generically correct for $\phi(x, a)$.
 - 4' By 16°e.ii, $g := \{k^{-1}(y(2n+2)) : n < \omega\}$ is $\overline{M}$ -generic for $k^{-1}(y(0))$.
 - 5' By 16°c, we have $\overline{M}[g] \models \phi[k^{-1}(y(1))^g, \overline{a}]$.
 - 6' By 16°e.iii, we have $k^{-1}(y(1))^g = x$.
 - 7' Thus, $\overline{M}[g] \models \phi[x, \overline{a}]$.
 - 8' By 18°3', we have $V \models \phi[x, a]$.

□

19° Analogously, $p[U] \subseteq \{x \in \mathbb{R} : \neg \phi(x, a)\}$.

20° **Claim.** For every $g \rightsquigarrow_{<\kappa} V$, we have

$$\{x \in \mathbb{R} : \phi(x, a)\} \subseteq p[T], \quad \{x \in \mathbb{R} : \neg \phi(x, a)\} \subseteq p[U]$$

inside $V[g]$.

Proof.

1' We may assume w.l.o.g. that $g \rightsquigarrow V^{\mathbb{P}}$ for some $\mathbb{P} \in H_\kappa$.

2' Let us work in $V[g]$ and let us show that

$$\{x \in \mathbb{R} : \phi(x, a)\} \subseteq p[T]$$

(the other inclusion is analogous).

3' Let $x \in \mathbb{R}$ be arbitrary such that $\phi(x, a)$. Let $\dot{x} \in V^{\mathbb{P}} \cap H_\kappa$ be such that $\dot{x}^g = x$.

4' Define $y \in H^\omega$ as follows:

- a. $y(0) := \mathbb{P}$
- b. $y(1) := \dot{x}$
- c. $y(2) \in g$ is such that $y(2) \Vdash \phi(\dot{x}, a)$
- d. $y(2n+3) := F(y \circ \mathbf{t}_n)$
- e. if $y(2n+3)$ is a dense subset of $\mathbb{P}$, then $y(2n+4)$ is an element of $y(2n+3) \cap g$ which is $\leq y(2n+2)$; otherwise, $y(2n+4) := y(2n+2)$.

5' We obviously have $(x, y) \in T$, and consequently $x \in p[T]$. □

21° By 18° and 19°, $p[T] \cap p[U] = \emptyset$ inside V , and consequently inside every generic extension of V .

22° This implies that in 20° the inclusions are in fact equalities. □

VI. Tree Production Lemma. Let λ be a limit of Woodin cardinals and let $\phi(x, a)$ be a formula with a parameter $a \in V$. Suppose that:

- a. for every $g \rightsquigarrow_{<\lambda} V$ and every $h \rightsquigarrow_{<\lambda} V[g]$, we have

$$(\forall x \in \mathbb{R} \cap V[g])(V[g] \models \phi[x, a] \iff V[g][h] \models \phi[x, a]);$$

- b. for every Woodin $\delta < \lambda$ and every generic $G \rightsquigarrow V^{\mathbb{Q}_\delta}$, we have

$$(\forall x \in \mathbb{R}^G)(V[G] \models \phi[x, a] \iff \text{Ult}(V, G) \models \phi[x, j_G(a)]).$$

Then $A := \{x \in \mathbb{R} : \phi(x, a)\}$ is λ -universally Baire and for every $g \rightsquigarrow_{<\lambda} V$, we have

$$A^g = \{x \in \mathbb{R}^g : V[g] \models \phi[x, a]\}.$$

Proof.

- 1° By 1.3-XXIV, it suffices to show that for every regular uncountable $\kappa < \lambda$, we have that A is κ -universally Baire and $A^g = \{x \in \mathbb{R}^g : V[g] \models \phi[x, a]\}$ for every $g \rightsquigarrow_{<\kappa} V$.

2° To that end, let us fix a regular $\kappa \in (\omega, \lambda)$, a transitive hull

$$\chi : \mathcal{H} \prec (V, \in, \kappa, a)$$

with $H_\kappa \subseteq H$ and $|H| < \lambda$, and let us show that χ is $<\kappa$ -generically correct for $\phi(x, a)$. By 2-V, this suffices for the conclusion.

3° Assume otherwise, that there is a stationary set $b \subseteq [H]^\omega$ such that for every $M \in b$, $M \prec \mathcal{H}$ and for its anti-collapse k , we have that $\chi \circ k$ is not $<\kappa$ -generically correct for $\phi(x, a)$.

4° Let $\delta \in (\kappa, \lambda)$ be Woodin such that $H \in V_\delta$ and let $G \rightsquigarrow V^{\mathbb{Q}_\delta}$ be a generic containing b . Let $j := j_G : V \prec \text{Ult}(V, G) =: \mathfrak{M}$.

5° Inside $\mathfrak{M}$, we have that $j(\chi) : j(\mathcal{H}) \prec \mathfrak{M}$ is a transitive hull.

6° for every $M \in j(b)$, $M \prec j(\mathcal{H})$ and for its anti-collapse k , we have that $j(\chi) \circ k$ is not $<j(\kappa)$ -generically correct for $\phi(x, j(a))$.

7° **Claim.**

- a. $j[H] \prec j(\mathcal{H})$ and it is countable in $\mathfrak{M}$.
- b. $j[H] \in j(b)$
- c. $j \upharpoonright H$ is the anti-collapse for $j[H]$.
- d. $j(\chi) \circ (j \upharpoonright H)$ is $<j(\kappa)$ -generically correct for $\phi(x, j(a))$.

Proof.

1' Only the last point is not obvious and we verify it here.

2' Let $g \in \mathfrak{M}$ be such that $g \rightsquigarrow_\kappa H$. We want to show that for every $x \in \mathbb{R} \cap H[g]$, we have

$$H[g] \models \phi[x, a^{\mathcal{H}}] \iff \mathfrak{M} \models \phi[x, j(a)].$$

3' For $x \in \mathbb{R} \cap H[g]$, we have the following:

$$\mathfrak{M} \models \phi[x, j(a)] \iff V[G] \models \phi[x, a] \tag{2}$$

$$\iff V[g] \models \phi[x, a] \tag{3}$$

$$\iff H[g] \models \phi[x, a^{\mathcal{H}}]. \tag{4}$$

The explanations are as follows:

(2) this is just VI.b;

(3) this follows from VI.a and from the facts that g is generic over V (since $H_\kappa \subseteq H$) and that $g \in V[G]$;

(4) this follows from the fact that $\chi : H \prec V$ is sufficiently elementary.

□

8° Points 6° and 7° are contradictory.

□

3 Determinacy in $L(\mathbb{R})$

- I. For $g \rightsquigarrow V^{\text{Col}(\omega, < \lambda)}$ and $\alpha < \lambda$, we will write $g_\alpha := g \restriction \alpha = g \cap \text{Col}(\omega, < \alpha)$.
- II. **Definition.** Let λ be a beth fixed point and let $g \rightsquigarrow V^{\text{Col}(\omega, < \lambda)}$. We define *the set of symmetric reals (w.r.t. g)* as

$$\mathbb{R}^* = \mathbb{R}^*[g] := \bigcup_{\alpha < \lambda} \mathbb{R}^{g_\alpha}.$$

- III. By $\mathbb{R}^*$ we shall also mean the appropriate name for the set of symmetric reals. We require of it to satisfy the following conditions.

- For every $\mathbb{P} \in H_\lambda$,

$$V^\mathbb{P} \models \text{Col}(\omega, < \lambda) \Vdash \ulcorner \mathbb{R}^* \text{ is the set of symmetric reals } \urcorner,$$

where $\mathbb{R}^*$ is seen as a name in $(V^\mathbb{P})^{\text{Col}(\omega, < \lambda)}$.

- For every $\alpha < \lambda$ and every $i \in \text{Aut}(\mathbb{B}(\text{Col}(\omega, < \alpha)))$, we have $i_*\mathbb{R}^* = \mathbb{R}^*$, where i is considered to be extended to $\mathbb{B}(\text{Col}(\omega, < \lambda))$ in the trivial manner.

- IV. **Proposition.** Let $g \rightsquigarrow V^{\text{Col}(\omega, < \lambda)}$. Then $V[g] \models \mathbb{R}^{L(\mathbb{R}^*)} = \mathbb{R}^*$.

Proof.

- 1° We only need to show $(\subseteq)$. We work in $V[g]$.
- 2° Let $x \in \mathbb{R}^{L(\mathbb{R}^*[g])}$ be arbitrary.
- 3° Then there are $\alpha < \text{Ord}$, $\zeta < \alpha$, $a \in \mathbb{R}^*$, and a formula ϕ such that for every $m, n < \omega$,

$$x(m) = n \iff L_\alpha(\mathbb{R}^*[g]) \models \phi[m, n, a, \zeta].$$

- 4° Let $\beta < \lambda$ be such that $a \in \mathbb{R}^{V[g_\beta]}$. Let $g^\beta \rightsquigarrow V[g_\beta]^{\text{Col}(\omega, < \lambda)}$ be such that $V[g_\beta][g^\beta] = V[g]$.

- 5° Then

$$x(m) = n \iff V[g_\beta][g^\beta] \models L_\alpha(\mathbb{R}^*[g^\beta]) \models \phi[m, n, a, \zeta] \quad (5)$$

$$\iff V[g_\beta] \models \text{Col}(\omega, < \lambda) \Vdash L_\alpha(\mathbb{R}^*) \models \phi[m, n, a, \zeta]. \quad (6)$$

Equivalence (6) follows from the fact that the automorphisms in

$$\bigcup_{\gamma < \lambda} \text{Aut}(\mathbb{B}(\text{Col}(\omega, < \gamma)))^{V[g_\beta]}$$

act transitively on $\mathbb{B}(\text{Col}(\omega, < \lambda))$ and the fact that all the names on the right hand side of the forcing symbol are invariant for these automorphisms.

- 6° We can now conclude that $x \in \mathbb{R}^{g_\beta} \subseteq \mathbb{R}^*[g]$.

□

V. **Definition.** A *successor Woodin cardinal* is a Woodin cardinal that is not a limit of Woodin cardinals. For a set b and an ordinal λ , we define $(b, \lambda)_{\text{sw}}$ to be the set of all successor Woodins δ such that $b \in V_\delta$ and $\delta < \lambda$.

VI. **Definition.** Let λ be a limit of Woodins. A *pre-generic for $\mathbb{Q}_\lambda$ over V* is a sequence $G = (G_\delta : \delta \in A)$ such that:

- a. $A = (\alpha, \lambda)_{\text{sw}}$ for some $\alpha < \lambda$;
- b. $G_\delta \rightsquigarrow V^{\mathbb{Q}_\delta}$ for every $\delta \in A$;
- c. $G_{\delta_1} = G_{\delta_2} \cap \mathbb{Q}_{\delta_1}$ for every $\delta_1 < \delta_2$ in A .

The ultrapower $\text{Ult}(V, G)$ is the direct limit of models $\text{Ult}(V, G_\delta)$ ($\delta \in A$), connected by the obvious embeddings.

VII. **Proposition.** Let Ω be large enough and let us work in $V^{\text{Col}(\omega, \Omega)}$. Suppose that:

- a. $V \models \ulcorner \lambda \text{ is a limit of Woodin cardinals} \urcorner$;
- b. $b \in \mathbb{Q}_\lambda^V$
- c. $\alpha \in \text{Ord}$
- d. $H \rightsquigarrow V^{\text{Col}(\omega, < \lambda)}$.

Then there is a pre-generic G for $\mathbb{Q}_\lambda$ over V such that:

- i. $b \in G_\delta$ for every $\delta \in \text{dom } G$;
- ii. $\alpha \in \text{wfp } \text{Ult}(V, G)$
- iii. $\mathbb{R}^{\text{Ult}(V, G)} = \mathbb{R}^*[H]$.

Proof. We need two auxiliary lemmas. The proof is given in [3-X](#). □

VIII. **Lemma.** Let M be a countable transitive model of set theory. Suppose that λ is a beth fixed point in M . Let $S \subseteq \mathbb{R}$ be such that:

- a. $|S| = \omega$
- b. every $x \in S$ is $< \lambda$ -generic over M ;
- c. for every $x, y \in S$, we have $x \oplus y \in S$;
- d. for every $y \in S$ and $x \in \mathbb{R}^{M[y]}$, we have $x \in S$;
- e. $\lambda = \sup\{\omega_1^{M[x]} : x \in S\}$.

Then there is $H \rightsquigarrow M^{\text{Col}(\omega, < \lambda)}$ such that $S = \mathbb{R}^*[H]$.

Proof.

- 1° Let $S = \{x_n : n < \omega\}$ and let $(D_n : n < \omega)$ enumerate all the dense subsets of $\text{Col}(\omega, < \lambda)$ that are in M . Let us also fix a sequence $(\gamma_n : n < \omega)$ cofinal in λ .
- 2° We define by recursion the following:
 - i. an increasing sequence of ordinals $(\alpha_n : n < \omega)$ such that $\alpha_n > \gamma_n$ for every $n < \omega$;
 - ii. an increasing sequence of generics $H_n \rightsquigarrow M^{\text{Col}(\omega, < \alpha_n)}$ with $H_n \cap D_n \neq \emptyset$ and $x_n \in M[H_n]$;
 - iii. a sequence of reals $h_n \in S$ such that $M[H_n] \subseteq M[h_n]$.
- 3° Once we have done this, $H := \bigcup \{H_n : n < \omega\}$ will obviously be as required. Namely:
 - by 2°i, we have $\lambda = \sup\{\alpha_n : n < \omega\}$;
 - by 2°ii, we have $H \rightsquigarrow M^{\text{Col}(\omega, < \lambda)}$;
 - again by 2°ii, we have $S \subseteq \mathbb{R}^*[H]$;
 - by 2°iii and d, we have $\mathbb{R}^*[H] \subseteq S$.
- 4° Let us proceed with the recursion. We first consider the stage $n = 0$.
- 5° Since x_0 is $< \lambda$ -generic over M , we can choose by e a real $h'_0 \in S$ such that in $M[x_0 \oplus h'_0]$ the relevant part of $M[x_0]$ is countable. By “the relevant part”, we mean the part whose countability permits the following conclusion: there is an $\alpha'_0 < \lambda$ and $H'_0 \in M[x_0 \oplus h'_0]$ with $H'_0 \rightsquigarrow M^{\text{Col}(\omega, < \alpha'_0)}$ and $x_0 \in M[H'_0]$.
- 6° There is some $p_0 \in D_0$ such that $p_0 \upharpoonright (\omega \times \alpha'_0) \in H'_0$.
- 7° Let $\alpha_0 \in (\max\{\alpha'_0, \gamma_0\}, \lambda)$ be such that $p_0 \in \text{Col}(\omega, < \alpha_0)$.
- 8° By e, we can choose $h''_0 \in S$ such that the relevant part of $M[x_0 \oplus h'_0]$ is countable.
- 9° Let $h_0 := x_0 \oplus h'_0 \oplus h''_0$. The countability of the relevant part now yields a generic $H_0 \rightsquigarrow M^{\text{Col}(\omega, < \alpha_0)}$ such that $H_0 \in M[h_0]$, $H_0 \cap \text{Col}(\omega, < \alpha'_0) = H'_0$, and $p_0 \in H_0$.
- 10° We now consider the recursive stage $n + 1$.
- 11° Since $x_{n+1} \oplus h_n \in S$ is $< \lambda$ -generic over M , there is a real $h'_{n+1} \in S$ such that in $M[x_{n+1} \oplus h_n \oplus h'_{n+1}]$ the relevant part of $M[x_{n+1} \oplus h_n]$ is countable.
- 12° Then there are $\alpha'_{n+1} \in (\alpha_n, \lambda)$ and $H'_{n+1} \in M[x_{n+1} \oplus h_n \oplus h'_{n+1}]$ with $H'_{n+1} \rightsquigarrow M^{\text{Col}(\omega, < \alpha'_{n+1})}$, $x_{n+1}, h_n \in M[H'_{n+1}]$, and $H'_{n+1} \cap \text{Col}(\omega, < \alpha_n) = H_n$.
- 13° There is some $p_{n+1} \in D_{n+1}$ such that $p_{n+1} \upharpoonright (\omega \times \alpha'_{n+1}) \in H'_{n+1}$.
- 14° Let $\alpha_{n+1} \in (\max\{\alpha'_{n+1}, \gamma_{n+1}\}, \lambda)$ be such that $p_{n+1} \in \text{Col}(\omega, < \alpha_{n+1})$.
- 15° By e, we can choose $h''_{n+1} \in S$ such that the relevant part of $M[x_{n+1} \oplus h_n \oplus h'_{n+1}]$ is countable in $M[x_{n+1} \oplus h_n \oplus h'_{n+1} \oplus h''_{n+1}]$.
- 16° Let $h_{n+1} := x_{n+1} \oplus h_n \oplus h'_{n+1} \oplus h''_{n+1} \in S$. There is a generic $H_{n+1} \rightsquigarrow M^{\text{Col}(\omega, < \alpha_{n+1})}$ such that $H_{n+1} \in M[h_{n+1}]$, $H_{n+1} \cap \text{Col}(\omega, < \alpha'_{n+1}) = H'_{n+1}$, and $p_{n+1} \in H_{n+1}$.

□

IX. Lemma. Suppose that λ is a limit of Woodins. Let a consist of all countable $M \prec H(\lambda^+)$ for which there exists $\alpha_0 < \lambda$ such that for every $\delta \in (\alpha_0, \lambda)_{\text{sw}} \cap M$, model M is self-generic with respect to $\mathbb{Q}_\delta$. Then a is projectively stationary³.

Proof.

1° Let $b \in \mathbb{Q}_\lambda$ and $F : H(\lambda^+)^{<\omega} \rightarrow H(\lambda^+)$ be arbitrary. We want to find $M \in a$ such that M is closed for F and $M \cap (\cup b) \in b$.

2° Let $\delta_0 \in (b, \lambda)_{\text{sw}}$ be the minimal.

3° We will construct a countable $N \prec H((2^\lambda)^+)$ such that:

- a. $F, \lambda \in N$
- b. for every $\delta \in (\delta_0, \lambda)_{\text{sw}} \cap N$, model M is self-generic with respect to $\mathbb{Q}_\delta$;
- c. $N \cap (\cup b) \in b$.

Then $M := N \cap H(\lambda^+)$ is as required.

4° We define a sequence of models $(N_\alpha : \alpha < \chi)$ and a sequence of ordinals $(\delta_\alpha : \alpha < \chi)$, for $\xi \leq \omega_1$, by recursion on α , such that:

- a. $N_\alpha \prec H((2^\lambda)^+)$ is countable;
- b. $N_\alpha \subseteq N_\beta$ for $\alpha < \beta < \chi$;
- c. δ_α is the α^{th} element of $(b, \lambda)_{\text{sw}} \cap N_\beta$ for $\alpha \leq \beta$;
- d. N_α and N_β agree up to $\sup(0, \delta_\alpha)_{\text{sw}}$ for $\alpha < \beta$;
- e. N_α is self-generic with respect to $\mathbb{Q}_{\delta_\beta}$ for every $\beta \leq \alpha$.

The value of χ will be determined by the recursion.

5° Let us consider the stage $\alpha = 0$.

6° There is a countable $N'_0 \prec (H((2^\lambda)^+), \in, b, \delta_0, \lambda, F)$ such that $N'_0 \cap (\cup b) \in b$.

7° By lemma 3-XII of [Kasum, 2021b], there is a countable $N_0 \prec H((2^\lambda)^+)$ such that:

- a. $N'_0 \subseteq N_0$
- b. N_0 and N'_0 agree up to $\text{rank}(b) + \omega$;
- c. N_0 is self-generic with respect to $\mathbb{Q}_{\delta_0}$.

8° Let us now consider the successor stage $\alpha + 1$.

9° Let $\delta_{\alpha+1} := \min(\delta_\alpha, \lambda)_{\text{sw}} \in N_\alpha$.

10° Since δ_α is the α^{th} element of $(b, \lambda)_{\text{sw}} \cap N_\alpha$, $\delta_{\alpha+1}$ is the $(\alpha + 1)^{\text{th}}$ element of that set.

11° By lemma 3-XII of [Kasum, 2021b], there is a countable $N_{\alpha+1} \prec H((2^\lambda)^+)$ such that:

- a. $N_\alpha \subseteq N_{\alpha+1}$

³I.e. $a \cap (b \upharpoonright H(\lambda^+))$ is stationary for every $b \in \mathbb{Q}_\lambda$

- b. N_α and $N_{\alpha+1}$ agree up to $\sup(0, \delta_{\alpha+1})_{\text{sw}} \in N_\alpha$;
- c. $N_{\alpha+1}$ is self-generic with respect to $\mathbb{Q}_{\delta_{\alpha+1}}$.
- 12° By 11°b, δ_β is the β^{th} element of $(b, \lambda)_{\text{sw}} \cap N_{\alpha+1}$ for every $\beta \leq \alpha + 1$.
- 13° By 11°b, the recursive hypothesis, and 11°c, we see tha $N_{\alpha+1}$ is self-generic with respect to $\mathbb{Q}_{\delta_\beta}$ for every $\beta \leq \alpha + 1$.
- 14° It remains to consider the limit stage $\alpha < \omega_1$.
- 15° Let $N_\alpha := \bigcup_{\beta < \alpha} N_\beta$ and let $\eta_\alpha := \sup\{\delta_\beta : \beta < \alpha\}$.
- 16° If $\eta_\alpha < \lambda$ and $N_\alpha \cap [\eta_\alpha, \lambda)$, then we define δ_α as the smallest successor Woodin in $N_\alpha \cap [\eta_\alpha, \lambda)$ and continue the recursion.
- 17° Otherwise, set $\chi := \alpha$ and halt the recursion.
- 18° **Claim.** $\chi < \omega_1$

Proof.

- 1' Assume otherwise and let $N := \bigcup_{\alpha < \omega_1} N_\alpha$.
- 2' For $\alpha < \omega_1$, we may define $f(\alpha) := \sup\{\beta < \omega_1 : \delta_\beta \in N_\alpha\} < \omega_1$.
- 3' Let $\alpha < \omega_1$ be a limit such that $f[\alpha] \subseteq \alpha$.
- 4' We have $\delta_\alpha \in N_\alpha$.
- 5' Since α is a limit, there is $\beta < \alpha$ such that $\delta_\alpha \in N_\beta$.
- 6' This implies $\alpha \leq f(\beta) < \alpha$, which is a contradiction.

□

- 19° Then χ is a limit and $N := N_\chi = \bigcup_{\alpha < \chi} N_\alpha$ is as required in 3° (property b follows from the fact that $\delta \in (b, \lambda)_{\text{sw}} \cap N$ implies $\delta < \eta_\chi$).

□

X. Proof of proposition 3-VII..

► We first show that there are some G and H such that:

- a. $H \rightsquigarrow V^{\text{Col}(\omega, < \lambda)}$
- b. G pre-generic for $\mathbb{Q}_\lambda^V$ over V ;
- c. $b \in G_\delta$ for every $\delta \in \text{dom } G$;
- d. $\alpha \in \text{wfp Ult}(V, G)$
- e. $\mathbb{R}^{\text{Ult}(V, G)} = \mathbb{R}^*[H]$.

- 1° Let a consist of all countable $M \prec H(\lambda^+)$ for which there exists $\alpha_0 < \lambda$ such that for every $\delta \in (\alpha_0, \lambda)_{\text{sw}} \cap M$, model M is self-generic with respect to $\mathbb{Q}_\delta$.
- 2° $a \cap (b \upharpoonright H(\lambda^+))$ is stationary by 3-IX.
- 3° Let $\theta \gg \Omega$ be such that H_θ is sufficiently elementary in V . Let $k : M \prec H_\theta$ be a countable hull such that $\text{im } k$ contains everything relevant and $\text{im } k \cap H(\lambda^+) \in a \cap (b \upharpoonright H(\lambda^+))$. We denote $\bar{\cdot} := k^{-1}(\cdot)$.

4° We will show that there are some G and H such that:

- a. $H \rightsquigarrow M^{\text{Col}(\omega, < \bar{\lambda})}$
- b. G pre-generic for $\mathbb{Q}_{\bar{\lambda}}^M$ over M ;
- c. $\bar{b} \in G_\delta$ for every $\delta \in \text{dom } G$;
- d. $\bar{\alpha} \in \text{wfp Ult}(M, G)$
- e. $\mathbb{R}^{\text{Ult}(M, G)} = \mathbb{R}^*[H]$.

Once we have done this, we will have such G and H in $M^{\text{Col}(\omega, \bar{\Omega})}$ by Σ_1^1 -absoluteness, and by elementarity, we will have the conclusion.

5° Let $\alpha_0 \in M$ be such that for every $\delta \in (k(\alpha_0), \lambda)_{\text{sw}}$, $\text{im } k$ is self-generic with respect to $\mathbb{Q}_\delta$.

6° Let G be derived from k , i.e.

$$(\forall \delta \in (\alpha_0, \bar{\lambda})_{\text{sw}}^M)(\forall c \in \mathbb{Q}_\delta^M)(c \in G_\delta \iff k[\cup c] \in k(c)).$$

7° 4°b follows from 5°, while 4°c follows from $\text{im } k \cap (\cup b) \in b$.

8° We have canonical realizability embeddings $\text{Ult}(M, G_\delta) \rightarrow H_\theta$ such that diagrams

$$\begin{array}{ccccc} & & H_\theta & & \\ & & \nwarrow & & \\ M & \longrightarrow & \text{Ult}(M, G_{\delta_1}) & \longrightarrow & \text{Ult}(M, G_{\delta_2}) \end{array}$$

commute for $\delta_1 < \delta_2$.

9° This means that $\text{Ult}(M, G)$ embeds in H_θ and it is consequently well-founded. In particular, 4°d holds.

10° Existence of H satisfying 4°a and 4°e follows from lemma 3-VIII.

► Let now $H_0 \rightsquigarrow V^{\text{Col}(\omega, < \lambda)}$ be given and let us show that there is G such that:

- a. G pre-generic for $\mathbb{Q}_\lambda^V$ over V ;
- b. $b \in G_\delta$ for every $\delta \in \text{dom } G$;
- c. $\alpha \in \text{wfp Ult}(V, G)$
- d. $\mathbb{R}^{\text{Ult}(V, G)} = \mathbb{R}^*[H_0]$.

11° Let $\phi(S, b, \alpha, \lambda)$ state: $\ulcorner$ there is G pre-generic for $\mathbb{Q}_\lambda^V$ over V such that:

- i. $b \in G_\delta$ for every $\delta \in \text{dom } G$;
- ii. $\alpha \in \text{wfp Ult}(V, G)$
- iii. $\mathbb{R}^{\text{Ult}(V, G)} = S^\ulcorner$.

We need to prove $V^{\text{Col}(\omega, \Omega)} \models \phi(\mathbb{R}^*[H_0], b, \alpha, \lambda)$

12° Let $H \rightsquigarrow V^{\text{Col}(\omega, < \lambda)}$ be the generic constructed above and let $\Delta \rightsquigarrow V[H]^{\text{Col}(\omega, \Omega)}$ be such that $V[\Gamma] = V[H][\Delta]$.

13° Then $V[H][\Delta] \models \phi(\mathbb{R}^*[H], b, \alpha, \lambda)$

14° There is some $p \in \text{Col}(\omega, \Omega)$ such that

$$V[H] \models (p \Vdash \phi(\mathbb{R}^*[H], b, \alpha, \lambda)).$$

15° Since $\text{Col}(\omega, \Omega)$ is homogeneous, we have

$$V[H] \models (\text{Col}(\omega, \Omega) \Vdash \phi(R^*[H], S, b, \alpha, \lambda)).$$

16° This means that for some $q \in \text{Col}(\omega, < \lambda)$, we have

$$V \models (q \Vdash (\text{Col}(\omega, \Omega) \Vdash \phi(\mathbb{R}^*, b, \alpha, \lambda))).$$

17° Since $\mathbb{R}^*$ is invariant for the automorphisms in

$$\bigcup_{\alpha < \lambda} \text{Aut}(\mathbb{B}(\text{Col}(\omega, < \alpha)))$$

and since those automorphisms act transitively on $\mathbb{B}(\text{Col}(\omega, < \lambda))^V$, we have

$$\text{Col}(\omega, < \lambda) \Vdash (\text{Col}(\omega, \Omega) \Vdash \phi(\mathbb{R}^*, b, \alpha, \lambda)).$$

18° Thus,

$$V[H_0] \models (\text{Col}(\omega, \Omega) \Vdash \phi(\mathbb{R}^*[H_0], b, \alpha, \lambda)).$$

19° There is $\Delta' \rightsquigarrow V[H_0]^{\text{Col}(\omega, \Omega)}$ such that $V[\Gamma] = V[H_0][\Delta']$. We have

$$V[\Gamma] = V[H_0][\Delta'] \models \phi(\mathbb{R}^*[H_0], b, \alpha, \lambda),$$

which was to be proved. □

XI. Theorem. Suppose that λ is a limit of Woodin cardinals. Then

$$V^{\text{Col}(\omega, < \lambda)} \models (L(\mathbb{R}^*) \models \text{AD}).$$

Proof.

1° Assume otherwise and let $H \rightsquigarrow V^{\text{Col}(\omega, < \lambda)}$ be such that $L(\mathbb{R}^*[H]) \models \neg \text{AD}$.

2° There are ordinals $\alpha < \zeta < \text{Ord}$, a formula ϕ , and a real $a \in \mathbb{R}^*[H]$ such that the set

$$\{x \in \mathbb{R}^*[H] : L_\alpha(\mathbb{R}^*[H]) \models \phi[x, \zeta, a]\}$$

is not determined in $L(\mathbb{R}^*)$.

3° $a \in \mathbb{R}^{g_\beta}$ for some $\beta < \lambda$. We may assume w.l.o.g. that $\beta = 0$, i.e. $a \in V$.

4° Let $\chi(\alpha, \phi, \zeta, a, S)$ be the formula “triple (α, ϕ, ζ) is the lexicographically minimal $\alpha < \zeta < \text{Ord}$, ϕ is a formula, and set

$$\{x \in S : L_\alpha(S) \models \phi[x, \zeta, a]\}$$

is not determined in $L(S)^\top$.

- 5° Let $\alpha_0, \phi_0, \zeta_0$ be such that $V[H] \models \chi(\alpha_0, \phi_0, \zeta_0, a, \mathbb{R}^*[H])$. We have $V[H] \models \ulcorner(\alpha_0, \phi_0, \zeta_0) \text{ is the unique solution for } \chi(\alpha, \phi, \zeta, a, \mathbb{R}^*)\urcorner$. (Intuitively, $(\alpha_0, \phi_0, \zeta_0)$ describe the minimal non-determined set.)
- 6° By the usual homogeneity argument, we have $V^{\text{Col}(\omega, < \lambda)} \models \ulcorner(\alpha_0, \phi_0, \zeta_0) \text{ is the unique solution for } \chi(\alpha, \phi, \zeta, a, \mathbb{R}^*)\urcorner$.
- 7° Let $\psi(x, a)$ denote formula

$$\text{Col}(\omega, < \lambda) \Vdash \forall \alpha \forall \phi \forall \zeta (\chi(\alpha, \phi, \zeta, a, \mathbb{R}^*) \implies L_\alpha(\mathbb{R}^*) \models \phi[x, \zeta, a]).$$

(Intuitively, ψ says that real x belongs to the minimal non-determined set in $L(\mathbb{R}^*)$.)

- 8° **Claim.** For every $g \rightsquigarrow_{< \lambda} V$, every $h \rightsquigarrow_{< \lambda} V[g]$, and every $x \in \mathbb{R}^g$, we have

$$V[g] \models \psi[x, a] \iff V[g][h] \models \psi[x, a].$$

(Intuitively, the minimal non-determined set in $L(\mathbb{R}^*)$ does not depend on the ground model.)

Proof.

- 1' Let $K \rightsquigarrow V[g][h]^{\text{Col}(\omega, < \lambda)}$ be arbitrary. Let $h \cdot K \rightsquigarrow V[g]^{\text{Col}(\omega, < \lambda)}$ correspond to $h * K$ and let $g \cdot h \cdot K \rightsquigarrow V^{\text{Col}(\omega, < \lambda)}$ correspond to $g * (h \cdot K)$, in some reasonable manner. We have

$$V[g][h][K] = V[g][h \cdot K] = V[g \cdot h \cdot K],$$

$$\mathbb{R}^*[K] = \mathbb{R}^*[h \cdot K] = \mathbb{R}^*[g \cdot h \cdot K] =: \mathbb{R}^*.$$

- 2' Suppose first that $V[g] \models \psi[x, a]$. We have

$$V[g][h \cdot K] \models \forall \alpha \forall \phi \forall \zeta (\chi(\alpha, \phi, \zeta, a, \mathbb{R}^*) \implies L_\alpha(\mathbb{R}^*) \models \phi[x, \zeta, a]).$$

- 3' Since $V[g][h \cdot K] = V[g][h][K]$, the usual homogeneity argument yields

$$V[g][h] \models \Vdash \forall \alpha \forall \phi \forall \zeta (\chi(\alpha, \phi, \zeta, a, \mathbb{R}^*) \implies L_\alpha(\mathbb{R}^*) \models \phi[x, \zeta, a]),$$

i.e. $V[g][h] \models \psi[x, a]$.

- 4' Suppose now that $V[g] \models \neg \psi[x, a]$, i.e.

$$V[g] \models \not\models \forall \alpha \forall \phi \forall \zeta (\chi(\alpha, \phi, \zeta, a, \mathbb{R}^*) \implies L_\alpha(\mathbb{R}^*) \models \phi[x, \zeta, a]).$$

- 5' Homogeneity yields

$$V[g] \models \Vdash \neg \forall \alpha \forall \phi \forall \zeta (\chi(\alpha, \phi, \zeta, a, \mathbb{R}^*) \implies L_\alpha(\mathbb{R}^*) \models \phi[x, \zeta, a]).$$

- 6' This means

$$V[g][h \cdot K] \models \neg \forall \alpha \forall \phi \forall \zeta (\chi(\alpha, \phi, \zeta, a, \mathbb{R}^*) \implies L_\alpha(\mathbb{R}^*) \models \phi[x, \zeta, a]).$$

7' Since $V[g][h \cdot K] = V[g][h][K]$, the usual homogeneity argument again yields

$$V[g][h] \models \vdash \neg \forall \alpha \forall \phi \forall \zeta (\chi(\alpha, \phi, \zeta, a, \mathbb{R}^*) \implies L_\alpha(\mathbb{R}^*) \models \phi[x, \zeta, a]).$$

In particular, $V[g][h] \models \neg \psi[x, a]$.

□

9° Let $\theta(x, a)$ denote formula

$$\forall \alpha \forall \phi \forall \zeta (\chi(\alpha, \phi, \zeta, a, \mathbb{R}) \implies L_\alpha(\mathbb{R}) \models \phi[x, \zeta, a]).$$

(Intuitively, θ says that x belongs to the minimal non-determined set in $L(\mathbb{R})$.)

10° **Claim.** For every $g \rightsquigarrow_{<\lambda} V$ and every $x \in \mathbb{R}^g$, we have

$$V[g] \models \psi[x, a] \iff V[g] \models \theta[x, a].$$

(Intuitively, the minimal non-determined set in $L(\mathbb{R})$ is equal to the intersection of the minimal non-determined set in $L(\mathbb{R}^*)$ with $V[g]$.)

Proof.

1' Suppose $V[g] \models \psi[x, a]$ and let $K \rightsquigarrow V[g]^{\text{Col}(\omega, <\lambda)}$. We have

$$V[g][K] \models \forall \alpha \forall \phi \forall \zeta (\chi(\alpha, \phi, \zeta, a, \mathbb{R}^*[K]) \implies L_\alpha(\mathbb{R}^*[K]) \models \phi[x, \zeta, a]).$$

2' Since $V[g][K]$ is a $\text{Col}(\omega, <\lambda)$ -extension of V whose symmetric reals are $\mathbb{R}^*[K] =: \mathbb{R}^*$, step 6° implies

$$V[g][K] \models L_{\alpha_0}(\mathbb{R}^*) \models \phi_0[x, \zeta_0, a].$$

3' By 3-VII, there is a pre-generic G for $\mathbb{Q}_\lambda$ over $V[g]$ such that $\alpha_0 \in \text{wfp Ult}(V[g], G)$ and $\mathbb{R} \cap \text{Ult}(V[g], G) = \mathbb{R}^*$.

4' By 10°2', we have

$$\text{Ult}(V[g], G) \models L_{\alpha_0}(\mathbb{R}) \models \phi_0[x, \zeta_0, a].$$

5' Also, $\text{Ult}(V[g], G) \models \ulcorner (\alpha_0, \phi_0, \zeta_0) \text{ is the unique solution for } \chi(\alpha, \phi, \zeta, a, \mathbb{R}) \urcorner$.

6' We conclude

$$\text{Ult}(V[g], G) \models \forall \alpha \forall \phi \forall \zeta (\chi(\alpha, \phi, \zeta, a, \mathbb{R}) \implies L_\alpha(\mathbb{R}) \models \phi[x, \zeta, a]),$$

i.e. $\text{Ult}(V[g], G) \models \theta[x, a]$.

7' By elementarity, $V[g] \models \theta[x, a]$.

8' On the other hand, suppose that $V[g] \models \neg \psi[x, a]$. We can then choose $K \rightsquigarrow V^{\text{Col}(\omega, <\lambda)}$ such that

$$V[g][K] \models \exists \alpha \exists \phi \exists \zeta (\chi(\alpha, \phi, \zeta, a, \mathbb{R}^*) \wedge L_\alpha(\mathbb{R}^*) \models \neg \phi[x, \zeta, a]).$$

9' As before, step 6° implies

$$V[g][K] \models L_{\alpha_0}(\mathbb{R}^*) \models \neg\phi_0[x, \zeta_0, a].$$

10' We again choose an appropriate pre-generic G for $\mathbb{Q}_\lambda$ over $V[g]$ and get

$$\text{Ult}(V[g], G) \models L_{\alpha_0}(\mathbb{R}) \models \neg\phi_0[x, \zeta_0, a],$$

which yields $\text{Ult}(V[g], G) \models \neg\theta[x, a]$.

11' By elementarity, $V[g] \models \neg\theta[x, a]$.

□

11° **Claim.** For every Woodin $\delta < \lambda$, for every $G \rightsquigarrow V^{\mathbb{Q}_\delta}$, and for every $x \in \mathbb{R}^G$, we have

$$V[G] \models \psi[x, a] \iff \text{Ult}(V, G) \models \psi[x, a].$$

Proof.

1' Suppose $V[G] \models \psi[x, a]$.

2' By 10°, we have $V[G] \models \theta[x, a]$.

3' By recalling the of θ , we conclude $L(\mathbb{R}^{V[G]}) \models \theta[x, a]$.

4' Since $\text{Ult}(V, G)$ is well-founded and $\mathbb{R}^{\text{Ult}(V, G)} = \mathbb{R}^{V[G]}$, we conclude $\text{Ult}(V, G) \models \theta[x, a]$.

5' By 10°, we have

$$V \models (\forall x \in \mathbb{R})(\psi(x, a) \iff \theta(x, a)).$$

6' By elementarity,

$$\text{Ult}(V, G) \models (\forall x \in \mathbb{R})(\psi(x, a) \iff \theta(x, a)).$$

7' 11°4' and 11°6' imply

$$\text{Ult}(V, G) \models \psi[x, a].$$

8' Suppose now $V[G] \models \neg\psi[x, a]$.

9' As before, $V[G] \models \neg\theta[x, a]$ and consequently, $\text{Ult}(V, G) \models \neg\theta[x, a]$.

10' 11°6' now yields $\text{Ult}(V, G) \models \neg\psi[x, a]$.

□

12° Claims 8° and 11° allow for the application of the Tree Production Lemma 2-VI. We conclude that $A := \{x \in \mathbb{R} : \psi(x, a)\}$ is λ -universally Baire and for every $g \rightsquigarrow_{<\lambda} V$,

$$A^g = \{x \in \mathbb{R}^g : V[g] \models \psi[x, a]\}.$$

13° By 1.4-IX, set A is determined, as witnessed by a strategy σ . We may assume w.l.o.g. that σ is a winning strategy for I.

- 14° Statement “ σ is a winning strategy for I in the game on A ” is a statement in the second-order arithmetic about σ (i.e. a real that codes it) and λ -universally Baire set A .
- 15° By 1.4-XI, we have that for every $g \rightsquigarrow_{<\lambda} V$, strategy σ is winning for I in the game on A^g inside $V[g]$.
- 16° Let us now work inside $V[H]$. Set

$$\{x \in \mathbb{R}^* : L_{\alpha_0}(\mathbb{R}^*) \models \phi_0[x, \zeta_0, a]\}$$

is not determined in $L(\mathbb{R}^*)$.

- 17° In particular, there is a play $x \in \mathbb{R}^*$ according to σ such that

$$L_{\alpha_0}(\mathbb{R}^*) \models \neg \phi_0[x, \zeta_0, a].$$

- 18° There is $\beta < \lambda$ such that $x \in \mathbb{R}^{H_\beta}$. By homogeneity, we have

$$V[H_\beta] \models \vdash \exists \alpha \exists \phi \exists \zeta (\chi(\alpha, \phi, \zeta, a, \mathbb{R}^*) \wedge L_\alpha(\mathbb{R}^*) \models \neg \phi[x, \zeta, a]).$$

- 19° In particular, $V[H_\beta] \models \neg \psi[x, a]$.

- 20° This means $x \notin A^{H_\beta}$, while x is played according to σ . This is in a contradiction with 15°.

□

XII. **Corollary.** The consistency strength of theory $\text{ZF} + \text{AD}$ is at most that of theory $\text{ZFC} + \ulcorner \text{there are } \omega \text{ many Woodin cardinals} \urcorner$.

XIII. We now work towards pulling AD from $L(\mathbb{R}^*)$ to $L(\mathbb{R})$.

XIV. **Lemma.** Suppose that W is an inner model and that κ is measurable in W . Let $A, B \in W \restriction \kappa$ and suppose that there is $j : L_\kappa(A) \prec L_\kappa(B)$. Then there is $k : L(A) \prec L(B)$ extending j .

Proof.

- 1° Inside W , let μ be a normal measure on κ .
- 2° Let $(W_\alpha, i_{\alpha, \beta} : \alpha < \beta < \text{Ord})$ be the iteration of W via μ . Let $\kappa_\alpha := i_{0, \alpha}(\kappa)$ and let $\mu_\alpha := i_{0, \alpha}(\mu)$ for $\alpha < \text{Ord}$.
- 3° $i_{\alpha, \beta} \restriction (W_\alpha \restriction \kappa_\alpha) : W_\alpha \restriction \kappa_\alpha \prec W_\beta \restriction \kappa_\beta$ is the identity ($\alpha \leq \beta$). In other words,

$$W_\alpha \restriction \kappa_\alpha \prec W_\beta \restriction \kappa_\beta.$$

- 4° This means that $L_{\kappa_\alpha}(A) \prec L_{\kappa_\beta}(A)$ and $L_{\kappa_\alpha}(B) \prec L_{\kappa_\beta}(B)$ for $\alpha \leq \beta$.

- 5° By recursion on α , we construct sequence $(j_\alpha : \alpha < \text{Ord})$ of embeddings such that diagram

$$\begin{array}{ccccccc}
L_{\kappa_0}(A) & \xrightarrow{\prec} & L_{\kappa_1}(A) & \xrightarrow{\prec} & \dots & \xrightarrow{\prec} & L_{\kappa_\alpha}(A) \xrightarrow{\prec} \dots \\
j_0 \downarrow \prec & & j_1 \downarrow \prec & & & & j_\alpha \downarrow \prec \\
L_{\kappa_0}(B) & \xrightarrow{\prec} & L_{\kappa_1}(B) & \xrightarrow{\prec} & \dots & \xrightarrow{\prec} & L_{\kappa_\alpha}(B) \xrightarrow{\prec} \dots
\end{array}$$

commutes.

6° For the base step, we have $j_0 := j$.

7° Let us consider recursive step $\alpha + 1$.

8° We have

$$L_{\kappa_{\alpha+1}}(A) = \{f/\mu_\alpha : f : \kappa_\alpha \rightarrow L_{\kappa_\alpha}(A), f \in W_\alpha\},$$

$$L_{\kappa_{\alpha+1}}(B) = \{f/\mu_\alpha : f : \kappa_\alpha \rightarrow L_{\kappa_\alpha}(B), f \in W_\alpha\}.$$

9° We define $j_{\alpha+1} : L_{\kappa_{\alpha+1}}(A) \prec L_{\kappa_{\alpha+1}}(B)$ by setting $j_{\alpha+1}(f/\mu_\alpha) = (j_\alpha \circ f)/\mu_\alpha$. It is a matter of routine to show that this mapping is correctly defined and elementary.

10° For limit step α , we have

$$L_{\kappa_\alpha}(A) = \bigcup_{\beta < \alpha} L_{\kappa_\beta}(A), \quad L_{\kappa_\alpha}(B) = \bigcup_{\beta < \alpha} L_{\kappa_\beta}(B)$$

and we set

$$j_\alpha := \bigcup_{\beta < \alpha} j_\beta.$$

11° This concludes the recursion.

12° Since

$$L(A) = \bigcup_{\alpha < \text{Ord}} L_{\kappa_\alpha}(A), \quad L(B) = \bigcup_{\alpha < \text{Ord}} L_{\kappa_\alpha}(B),$$

we are done if we take

$$k := \bigcup_{\alpha < \text{Ord}} j_\alpha.$$

□

XV. Theorem. We work in $V^{\text{Col}(\omega, \Omega)}$ for Ω large enough. Suppose that:

- a. $V \models \ulcorner \lambda \text{ is a limit of Woodin cardinals} \urcorner$;
- b. $V \models \ulcorner \text{there is a measurable cardinal above } \lambda \urcorner$;
- c. $H \rightsquigarrow V^{\text{Col}(\omega, < \lambda)}$.

Then there is an elementary embedding $j : L(\mathbb{R}^V) \prec L(\mathbb{R}^*[H])$.

Proof.

- 1° Let $\kappa > \lambda$ be measurable.
- 2° Let G be a pre-generic for $\mathbb{Q}_\lambda$ over V such that $\kappa \in \mathbf{wfp Ult}(V, G)$ and $\mathbb{R}^{\mathbf{Ult}(V, G)} = \mathbb{R}^*[H]$.
- 3° $L_\kappa(\mathbb{R})^{\mathbf{Ult}(V, G)} = L_\kappa(\mathbb{R}^*[H])$
- 4° Thus, $j_G \upharpoonright L_\kappa(\mathbb{R}^V) : L_\kappa(\mathbb{R}^V) \prec L_\kappa(\mathbb{R}^*[H])$.
- 5° By lemma 3-XIV, there is an embedding $j : L(\mathbb{R}^V) \prec L(\mathbb{R}^*[H])$.

□

XVI. **Corollary.** If there are ω many Woodin cardinals with a measurable above them, then $L(\mathbb{R}) \models \text{AD}$.

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